$$\begin{aligned} \text{Maximize} \quad & 2x + 7y \\ \text{subject to} \quad & 3x + 5y \leq 15 \\ & 7x + 3y \leq 21 \\ & x \geq 0,\ y \geq 0 \end{aligned}$$

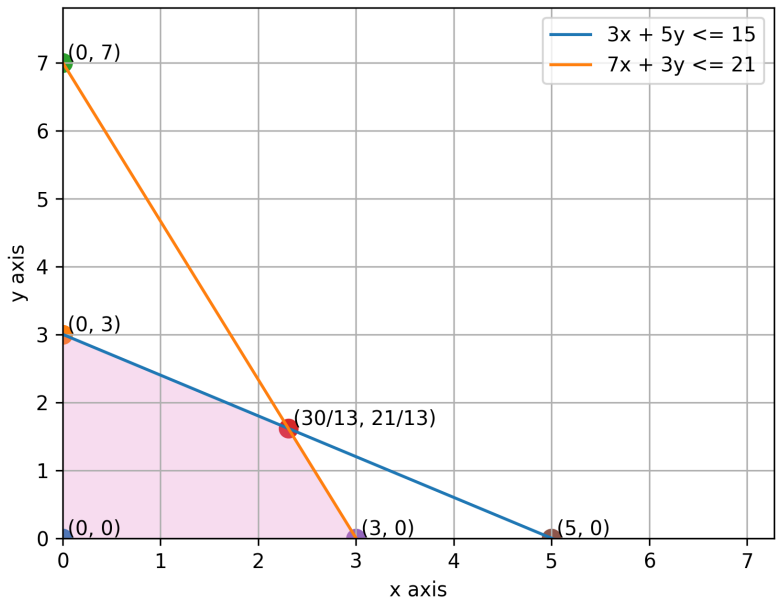
Gauss Elimination:
Echelon Form:

$$\left(\begin{array}{cc|c} 3 & 5 & 15 \\ 0 & -\frac{26}{3} & -14 \end{array}\right)$$

$$\left(\begin{array}{cc|c} 3 & 5 & 15 \\ 0 & -\frac{26}{3} & -14 \end{array}\right)$$

$$x = \frac{30}{13}$$

$$y = \frac{21}{13}$$



$$Z = 21$$

$$x = 0$$

$$y = 3$$

$$\begin{aligned} \text{Maximize} \quad & 3x + 5y \\ \text{subject to} \quad & 4x + 3y \leq 12 \\ & 5x + 4y \geq 20 \\ & x \geq 0,\ y \geq 0 \end{aligned}$$

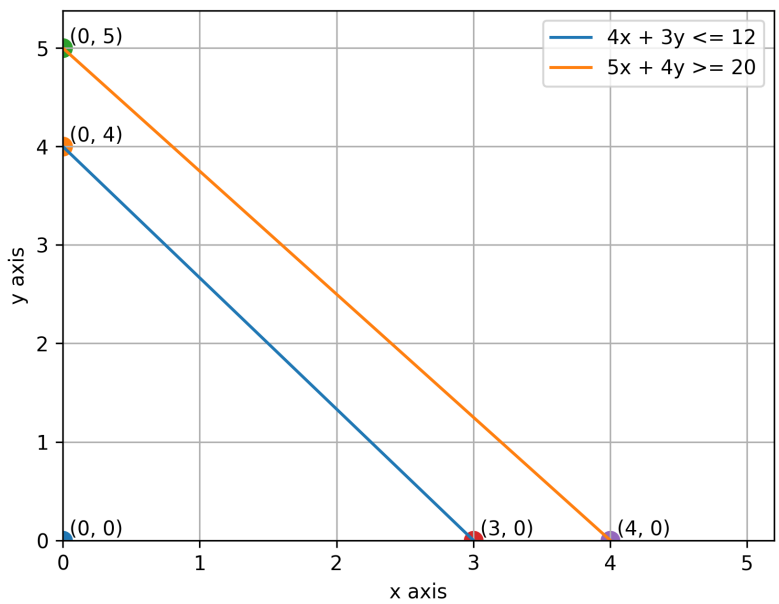
Gauss Elimination:
Echelon Form:

$$\left(\begin{array}{cc|c} 4 & 3 & 12 \\ 0 & \frac{1}{4} & 5 \end{array}\right)$$

$$\left(\begin{array}{cc|c} 4 & 3 & 12 \\ 0 & \frac{1}{4} & 5 \end{array}\right)$$

$$x = -12$$

$$y = 20$$



Infeasible Solution

$$\begin{aligned} \text{Maximize} \quad & x + 2y \\ \text{subject to} \quad & 3x + 2y \leq 6 \\ & 2x + 5y \geq 10 \\ & x \geq 0,\ y \geq 0 \end{aligned}$$

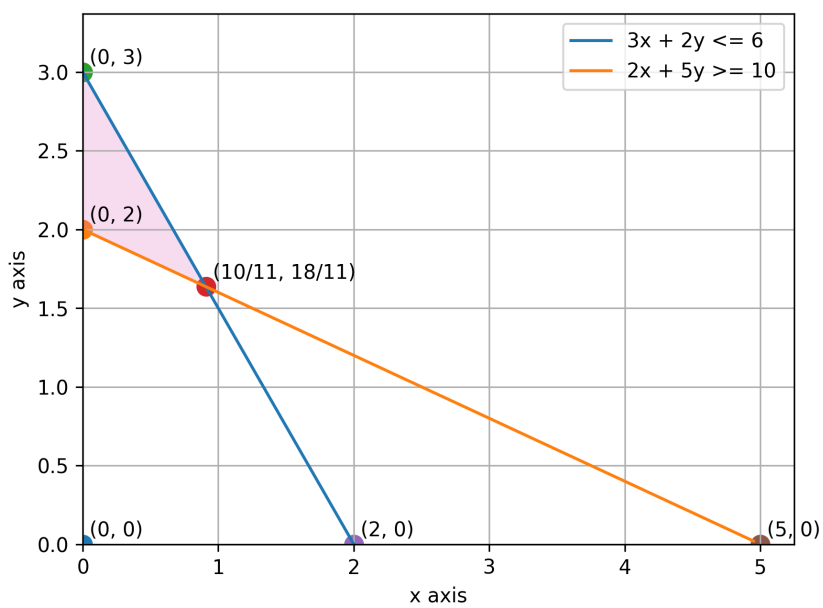
Gauss Elimination:
Echelon Form:

$$\left(\begin{array}{cc|c} 3 & 2 & 6 \\ 0 & \frac{11}{3} & 6 \end{array}\right)$$

$$\left(\begin{array}{cc|c} 3 & 2 & 6 \\ 0 & \frac{11}{3} & 6 \end{array}\right)$$

$$x = \frac{10}{11}$$

$$y = \frac{18}{11}$$



$$Z = 6$$

$$x = 0$$

$$y = 3$$

$$\begin{aligned} \text{Minimize} \quad & 3x - 10y \\ \text{subject to} \quad & 5x + 2y \geq 10 \\ & 4x + 3y \leq 12 \\ & x \geq 0,\ y \geq 0 \end{aligned}$$

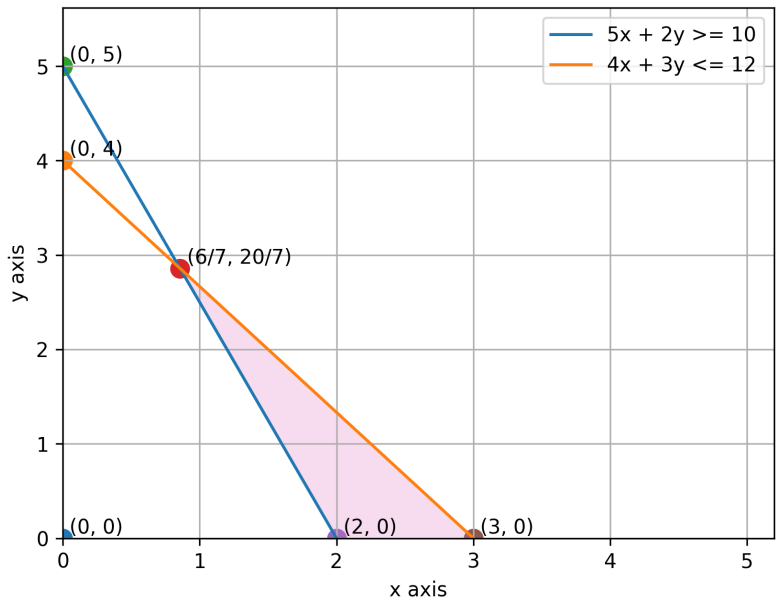
Gauss Elimination:
Echelon Form:

$$\left(\begin{array}{cc|c} 5 & 2 & 10 \\ 0 & \frac{7}{5} & 4 \end{array}\right)$$

$$\left(\begin{array}{cc|c} 5 & 2 & 10 \\ 0 & \frac{7}{5} & 4 \end{array}\right)$$

$$x = \frac{6}{7}$$

$$y = \frac{20}{7}$$



$$Z = -26$$

$$x = \frac{6}{7}$$

$$y = \frac{20}{7}$$

$$\begin{aligned} \text{Maximize} \quad & 5x + 4y \\ \text{subject to} \quad & 2x + 5y \geq 10 \\ & 3x + 4y \geq 12 \\ & x \geq 0,\ y \geq 0 \end{aligned}$$

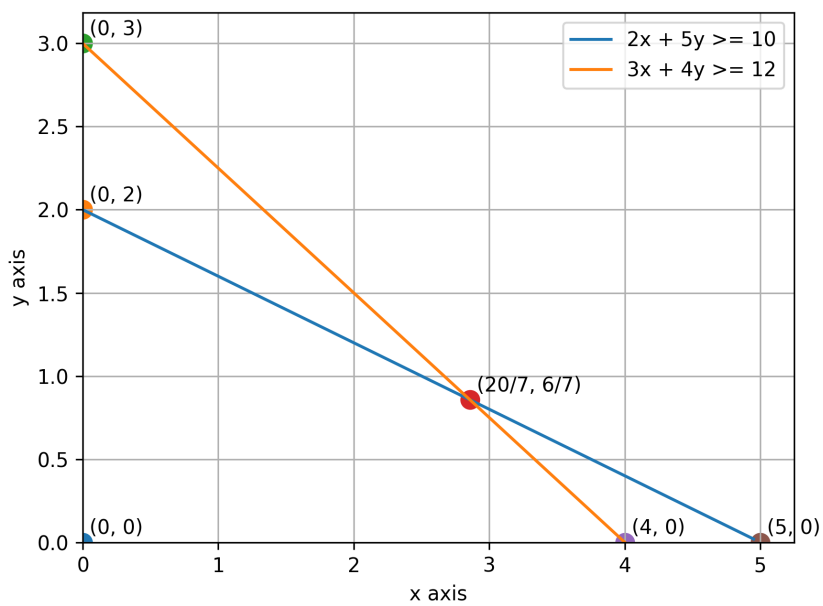
Gauss Elimination:
Echelon Form:

$$\left(\begin{array}{cc|c} 2 & 5 & 10 \\ 0 & -\frac{7}{2} & -3 \end{array}\right)$$

$$\left(\begin{array}{cc|c} 2 & 5 & 10 \\ 0 & -\frac{7}{2} & -3 \end{array}\right)$$

$$x = \frac{20}{7}$$

$$y = \frac{6}{7}$$



Unbounded Solution

$$\begin{aligned} \text{Minimize} \quad & 5x + 4y \\ \text{subject to} \quad & 2x + 5y \geq 10 \\ & 3x + 4y \geq 12 \\ & x \geq 0,\ y \geq 0 \end{aligned}$$

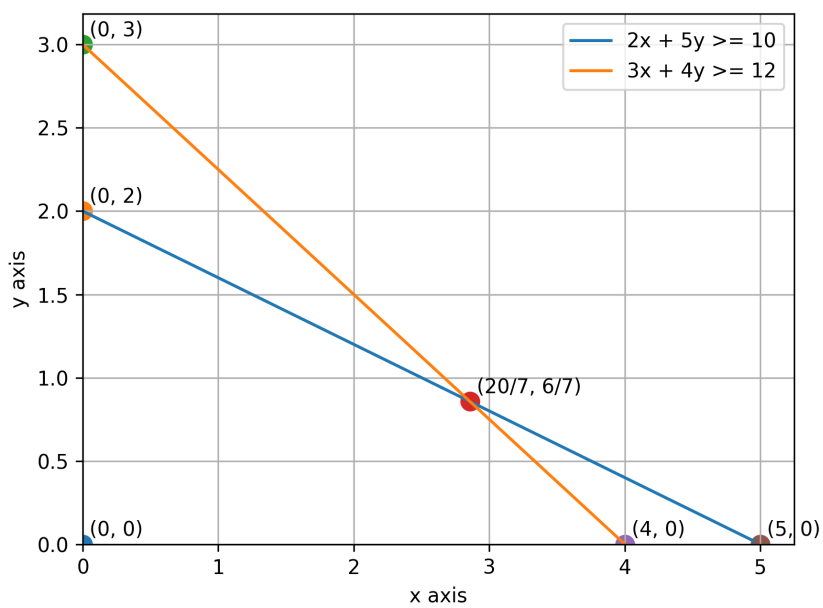
Gauss Elimination:
Echelon Form:

$$\left(\begin{array}{cc|c} 2 & 5 & 10 \\ 0 & -\frac{7}{2} & -3 \end{array}\right)$$

$$\left(\begin{array}{cc|c} 2 & 5 & 10 \\ 0 & -\frac{7}{2} & -3 \end{array}\right)$$

$$x = \frac{20}{7}$$

$$y = \frac{6}{7}$$



$$Z = 12$$

$$x = 0$$

$$y = 3$$

$$\begin{aligned} \text{Maximize} \quad & 10x + 4y \\ \text{subject to} \quad & 5x + 2y \leq 100 \\ & 3x + 2y \leq 90 \\ & x + 2y \leq 50 \\ & x \geq 0,\ y \geq 0 \end{aligned}$$

Gauss Elimination:
Echelon Form:

$$\left(\begin{array}{cc|c} 5 & 2 & 100 \\ 0 & \frac{4}{5} & 30 \end{array}\right)$$

$$\left(\begin{array}{cc|c} 5 & 2 & 100 \\ 0 & \frac{4}{5} & 30 \end{array}\right)$$

$$x = 5$$

$$y = \frac{75}{2}$$

Gauss Elimination:
Echelon Form:

$$\left(\begin{array}{cc|c} 5 & 2 & 100 \\ 0 & \frac{8}{5} & 30 \end{array}\right)$$

$$\left(\begin{array}{cc|c} 5 & 2 & 100 \\ 0 & \frac{8}{5} & 30 \end{array}\right)$$

$$x = \frac{25}{2}$$

$$y = \frac{75}{4}$$

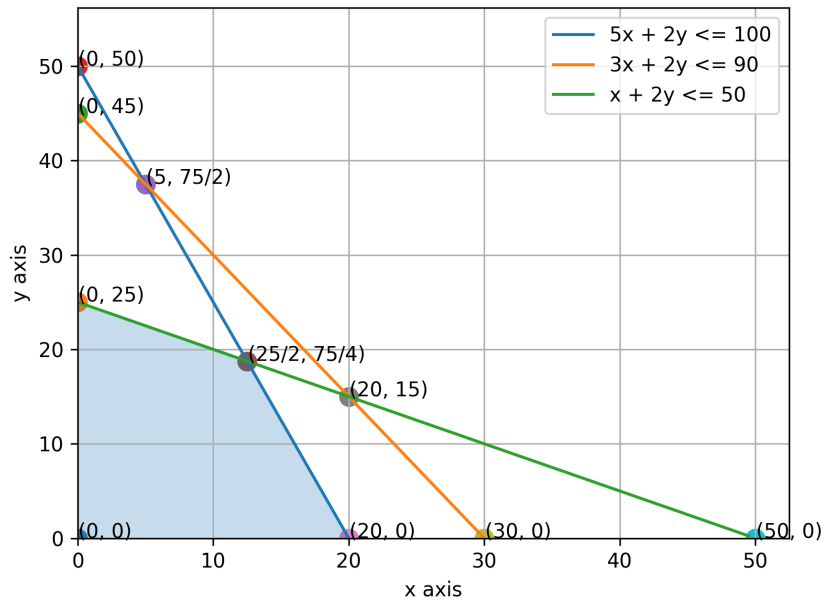
Gauss Elimination:
Echelon Form:

$$\left(\begin{array}{cc|c} 3 & 2 & 90 \\ 0 & \frac{4}{3} & 20 \end{array}\right)$$

$$\left(\begin{array}{cc|c} 3 & 2 & 90 \\ 0 & \frac{4}{3} & 20 \end{array}\right)$$

$$x = 20$$

$$y = 15$$



Infinitely Many Solutions

$$\begin{aligned} \text{Maximize} \quad & 3x + 2y \\ \text{subject to} \quad & x + y \leq 4 \\ & x - y \leq 2 \\ & x \geq 0,\ y \geq 0 \end{aligned}$$

Standard Form:

$$\begin{aligned} \text{Maximize} \quad & 3x + 2y \\ \text{subject to} \quad & s_1 + x + y = 4 \\ & s_2 + x - y = 2 \\ & x \geq 0,\ y \geq 0 \end{aligned}$$

			0	0	3	2	
BV	C	B	s_1	s_2	x	y	MR
s_1	0	4	1	0	1	1	4
s_2	0	2	0	1	1	-1	2
		$Z_j - C_j$	0	0	-3	-2	

			0	0	3	2	
BV	C	B	s_1	s_2	x	y	MR
s_1	0	2	1	-1	0	2	1
x	3	2	0	1	1	-1	$-$
		$Z_j - C_j$	0	3	0	-5	

			0	0	3	2	
BV	C	B	s_1	s_2	x	y	MR
y	2	1	$\frac{1}{2}$	$-\frac{1}{2}$	0	1	1
x	3	3	$\frac{1}{2}$	$\frac{1}{2}$	1	0	$-$
		$Z_j - C_j$	$\frac{5}{2}$	$\frac{1}{2}$	0	0	

$$Z = 11$$

$$x = 3$$

$$y = 1$$

$$\begin{aligned} \text{Maximize} \quad & x_1 + x_2 + 3x_3 \\ \text{subject to} \quad & 3x_1 + 2x_2 + x_3 \leq 3 \\ & 2x_1 + x_2 + 2x_3 \leq 2 \\ & x_1 \geq 0,\ x_2 \geq 0,\ x_3 \geq 0 \end{aligned}$$

Standard Form:

$$\begin{aligned} \text{Maximize} \quad & x_1 + x_2 + 3x_3 \\ \text{subject to} \quad & s_1 + 3x_1 + 2x_2 + x_3 = 3 \\ & s_2 + 2x_1 + x_2 + 2x_3 = 2 \\ & x_1 \geq 0,\ x_2 \geq 0,\ x_3 \geq 0 \end{aligned}$$

			0	0	1	1	3	
BV	C	B	s_1	s_2	x_1	x_2	x_3	MR
s_1	0	3	1	0	3	2	1	3
s_2	0	2	0	1	2	1	2	1
		$Z_j - C_j$	0	0	-1	-1	-3	

			0	0	1	1	3	
BV	C	B	s_1	s_2	x_1	x_2	x_3	MR
s_1	0	2	1	$-\frac{1}{2}$	2	$\frac{3}{2}$	0	3
x_3	3	1	0	$\frac{1}{2}$	1	$\frac{1}{2}$	1	1
		$Z_j - C_j$	0	$\frac{3}{2}$	2	$\frac{1}{2}$	0	

$$Z = 3$$

$$x_1 = 0$$

$$x_2 = 0$$

$$x_3 = 1$$

$$\begin{aligned} \text{Minimize} \quad & x_1 - 3x_2 + 2x_3 \\ \text{subject to} \quad & 3x_1 - x_2 + 2x_3 \leq 7 \\ & -2x_1 + 4x_2 \leq 12 \\ & -4x_1 + 3x_2 + 8x_3 \leq 10 \\ & x_1 \geq 0,\ x_2 \geq 0,\ x_3 \geq 0 \end{aligned}$$

Standard Form:

$$\begin{aligned} \text{Maximize} \quad & -x_1 + 3x_2 - 2x_3 \\ \text{subject to} \quad & s_1 + 3x_1 - x_2 + 2x_3 = 7 \\ & s_2 - 2x_1 + 4x_2 = 12 \\ & s_3 - 4x_1 + 3x_2 + 8x_3 = 10 \\ & x_1 \geq 0,\ x_2 \geq 0,\ x_3 \geq 0 \end{aligned}$$

			0	0	0	-1	3	-2	
BV	C	B	s_1	s_2	s_3	x_1	x_2	x_3	MR
s_1	0	7	1	0	0	3	-1	2	–
s_2	0	12	0	1	0	-2	4	0	3
s_3	0	10	0	0	1	-4	3	8	$\frac{10}{3}$
		$Z_j - C_j$	0	0	0	1	-3	2	

			0	0	0	-1	3	-2	
BV	C	B	s_1	s_2	s_3	x_1	x_2	x_3	MR
s_1	0	10	1	$\frac{1}{4}$	0	$\frac{5}{2}$	0	2	4
x_2	3	3	0	$\frac{1}{4}$	0	$-\frac{1}{2}$	1	0	–
s_3	0	1	0	$-\frac{3}{4}$	1	$-\frac{5}{2}$	0	8	–
		$Z_j - C_j$	0	$\frac{3}{4}$	0	$-\frac{1}{2}$	0	2	

			0	0	0	-1	3	-2	
BV	C	B	s_1	s_2	s_3	x_1	x_2	x_3	MR
x_1	-1	4	$\frac{2}{5}$	$\frac{1}{10}$	0	1	0	$\frac{4}{5}$	4
x_2	3	5	$\frac{1}{5}$	$\frac{3}{10}$	0	0	1	$\frac{2}{5}$	–
s_3	0	11	1	$-\frac{1}{2}$	1	0	0	10	–
		$Z_j - C_j$	$\frac{1}{5}$	$\frac{4}{5}$	0	0	0	$\frac{12}{5}$	

$$Z = 11$$

$$x_1 = 4$$

$$x_2 = 5$$

$$x_3 = 0$$

$$\begin{aligned} \text{Maximize} \quad & 2x + y \\ \text{subject to} \quad & 4x + 3y \leq 12 \\ & 4x + y \leq 8 \\ & 4x - y \leq 8 \\ & x \geq 0,\ y \geq 0 \end{aligned}$$

Standard Form:

$$\begin{aligned} \text{Maximize} \quad & 2x + y \\ \text{subject to} \quad & s_1 + 4x + 3y = 12 \\ & s_2 + 4x + y = 8 \\ & s_3 + 4x - y = 8 \\ & x \geq 0,\ y \geq 0 \end{aligned}$$

			0	0	0	2	1	
BV	C	B	s_1	s_2	s_3	x	y	MR
s_1	0	12	1	0	0	4	3	3
s_2	0	8	0	1	0	4	1	2
s_3	0	8	0	0	1	4	-1	2
		$Z_j - C_j$	0	0	0	-2	-1	

			0	0	0	2	1	
BV	C	B	s_1	s_2	s_3	x	y	MR
s_1	0	4	1	0	-1	0	4	1
s_2	0	0	0	1	-1	0	2	0
x	2	2	0	0	$\frac{1}{4}$	1	$-\frac{1}{4}$	$-$
		$Z_j - C_j$	0	0	$\frac{1}{2}$	0	$-\frac{3}{2}$	

			0	0	0	2	1	
BV	C	B	s_1	s_2	s_3	x	y	MR
s_1	0	4	1	-2	1	0	0	4
y	1	0	0	$\frac{1}{2}$	$-\frac{1}{2}$	0	1	$-$
x	2	2	0	$\frac{1}{8}$	$\frac{1}{8}$	1	0	16
		$Z_j - C_j$	0	$\frac{3}{4}$	$-\frac{1}{4}$	0	0	

			0	0	0	2	1	
BV	C	B	s_1	s_2	s_3	x	y	MR
s_3	0	4	1	-2	1	0	0	4
y	1	2	$\frac{1}{2}$	$-\frac{1}{2}$	0	0	1	$-$
x	2	$\frac{3}{2}$	$-\frac{1}{8}$	$\frac{3}{8}$	0	1	0	16
		$Z_j - C_j$	$\frac{1}{4}$	$\frac{1}{4}$	0	0	0	

$$Z = 5$$

$$x = \frac{3}{2}$$

$$y = 2$$

$$\begin{aligned} \text{Maximize} \quad & 2x + y \\ \text{subject to} \quad & x - y \leq 10 \\ & 2x - y \leq 40 \\ & x \geq 0,\ y \geq 0 \end{aligned}$$

Standard Form:

$$\begin{aligned} \text{Maximize} \quad & 2x + y \\ \text{subject to} \quad & s_1 + x - y = 10 \\ & s_2 + 2x - y = 40 \\ & x \geq 0,\ y \geq 0 \end{aligned}$$

			0	0	2	1	
BV	C	B	s_1	s_2	x	y	MR
s_1	0	10	1	0	1	-1	10
s_2	0	40	0	1	2	-1	20
		$Z_j - C_j$	0	0	-2	-1	

			0	0	2	1	
BV	C	B	s_1	s_2	x	y	MR
x	2	10	1	0	1	-1	–
s_2	0	20	-2	1	0	1	20
		$Z_j - C_j$	2	0	0	-3	

			0	0	2	1	
BV	C	B	s_1	s_2	x	y	MR
x	2	30	-1	1	1	0	–
y	1	20	-2	1	0	1	–
		$Z_j - C_j$	-4	3	0	0	

Unbounded Solution

$$\begin{aligned} \text{Maximize} \quad & 3x + 2y \\ \text{subject to} \quad & x - y \leq 1 \\ & 3x - 2y \leq 6 \\ & x \geq 0,\ y \geq 0 \end{aligned}$$

Standard Form:

$$\begin{aligned} \text{Maximize} \quad & 3x + 2y \\ \text{subject to} \quad & s_1 + x - y = 1 \\ & s_2 + 3x - 2y = 6 \\ & x \geq 0,\ y \geq 0 \end{aligned}$$

			0	0	3	2	
BV	C	B	s_1	s_2	x	y	MR
s_1	0	1	1	0	1	-1	1
s_2	0	6	0	1	3	-2	2
		$Z_j - C_j$	0	0	-3	-2	

			0	0	3	2	
BV	C	B	s_1	s_2	x	y	MR
x	3	1	1	0	1	-1	–
s_2	0	3	-3	1	0	1	3
		$Z_j - C_j$	3	0	0	-5	

			0	0	3	2	
BV	C	B	s_1	s_2	x	y	MR
x	3	4	-2	1	1	0	–
y	2	3	-3	1	0	1	–
		$Z_j - C_j$	-12	5	0	0	

Unbounded Solution

$$\begin{aligned} \text{Maximize} \quad & x_1 + 2x_2 + x_3 \\ \text{subject to} \quad & 2x_1 + x_2 - x_3 \geq -2 \\ & -2x_1 + x_2 - 5x_3 \leq 6 \\ & 4x_1 + x_2 + x_3 \leq 6 \\ & x_1 \geq 0,\ x_2 \geq 0,\ x_3 \geq 0 \end{aligned}$$

Standard Form:

$$\begin{aligned} \text{Maximize} \quad & x_1 + 2x_2 + x_3 \\ \text{subject to} \quad & s_1 - 2x_1 - x_2 + x_3 = 2 \\ & s_2 - 2x_1 + x_2 - 5x_3 = 6 \\ & s_3 + 4x_1 + x_2 + x_3 = 6 \\ & x_1 \geq 0,\ x_2 \geq 0,\ x_3 \geq 0 \end{aligned}$$

			0	0	0	1	2	1	
BV	C	B	s_1	s_2	s_3	x_1	x_2	x_3	MR
s_1	0	2	1	0	0	-2	-1	1	–
s_2	0	6	0	1	0	-2	1	-5	6
s_3	0	6	0	0	1	4	1	1	6
		$Z_j - C_j$	0	0	0	-1	-2	-1	

			0	0	0	1	2	1	
BV	C	B	s_1	s_2	s_3	x_1	x_2	x_3	MR
s_1	0	8	1	0	1	2	0	2	–
s_2	0	0	0	1	-1	-6	0	-6	1
x_2	2	6	0	0	1	4	1	1	0
		$Z_j - C_j$	0	0	2	7	0	1	

$$Z = 12$$

$$x_1 = 0$$

$$x_2 = 6$$

$$x_3 = 0$$

$$\begin{aligned} \text{Maximize} \quad & -4x_1 - x_2 \\ \text{subject to} \quad & 3x_1 + x_2 = 3 \\ & 4x_1 + 3x_2 \geq 6 \\ & x_1 + 2x_2 \leq 3 \\ & x_1 \geq 0,\ x_2 \geq 0 \end{aligned}$$

Standard Form:

$$\begin{aligned} \text{Maximize} \quad & -4x_1 - x_2 \\ \text{subject to} \quad & 3x_1 + x_2 = 3 \\ & -s_1 + 4x_1 + 3x_2 = 6 \\ & s_2 + x_1 + 2x_2 = 3 \\ & x_1 \geq 0,\ x_2 \geq 0 \end{aligned}$$

			$-M$	$-M$	0	0	-4	-1	
BV	C	B	A_1	A_2	s_1	s_2	x_1	x_2	MR
A_1	$-M$	3	1	0	0	0	3	1	1
A_2	$-M$	6	0	1	-1	0	4	3	$\frac{3}{2}$
s_2	0	3	0	0	0	1	1	2	3
		$Z_j - C_j$	0	0	M	0	$-7M+4$	$-4M+1$	

			$-M$	0	0	-4	-1	
BV	C	B	A_2	s_1	s_2	x_1	x_2	MR
x_1	-4	1	0	0	0	1	$\frac{1}{3}$	3
A_2	$-M$	2	1	-1	0	0	$\frac{5}{3}$	$\frac{6}{5}$
s_2	0	2	0	0	1	0	$\frac{5}{3}$	$\frac{6}{5}$
		$Z_j - C_j$	0	M	0	0	$\frac{-5M}{3} - \frac{1}{3}$	

			0	0	-4	-1	
BV	C	B	s_1	s_2	x_1	x_2	MR
x_1	-4	$\frac{3}{5}$	$\frac{1}{5}$	0	1	0	3
x_2	-1	$\frac{6}{5}$	$-\frac{3}{5}$	0	0	1	–
s_2	0	0	1	1	0	0	0
		$Z_j - C_j$	$-\frac{1}{5}$	0	0	0	

			0	0	-4	-1	
BV	C	B	s_1	s_2	x_1	x_2	MR
x_1	-4	$\frac{3}{5}$	0	$-\frac{1}{5}$	1	0	3
x_2	-1	$\frac{6}{5}$	0	$\frac{3}{5}$	0	1	–
s_1	0	0	1	1	0	0	0
		$Z_j - C_j$	0	$\frac{1}{5}$	0	0	

$$Z = -\frac{18}{5}$$

$$x_1 = \frac{3}{5}$$

$$x_2 = \frac{6}{5}$$

$$\begin{aligned} \text{Maximize} \quad & -x - y \\ \text{subject to} \quad & 3x + 2y \geq 30 \\ & -2x + 3y \leq -30 \\ & x + y \leq 5 \\ & x \geq 0,\ y \geq 0 \end{aligned}$$

Standard Form:

$$\begin{aligned} \text{Maximize} \quad & -x - y \\ \text{subject to} \quad & -s_1 + 3x + 2y = 30 \\ & -s_2 + 2x - 3y = 30 \\ & s_3 + x + y = 5 \\ & x \geq 0,\ y \geq 0 \end{aligned}$$

			$-M$	$-M$	0	0	0	-1	-1	
BV	C	B	A_1	A_2	s_1	s_2	s_3	x	y	MR
A_1	$-M$	30	1	0	-1	0	0	3	2	10
A_2	$-M$	30	0	1	0	-1	0	2	-3	15
s_3	0	5	0	0	0	0	1	1	1	5
		$Z_j - C_j$	0	0	M	M	0	$-5M+1$	$M+1$	

Infeasible Solution

$$\begin{aligned} \text{Maximize} \quad & -4x - y \\ \text{subject to} \quad & 3x + y = 3 \\ & 4x + 3y \geq 6 \\ & x + 2y \leq 3 \\ & x \geq 0,\ y \geq 0 \end{aligned}$$

Standard Form:

$$\begin{aligned} \text{Maximize} \quad & -4x - y \\ \text{subject to} \quad & 3x + y = 3 \\ & -s_1 + 4x + 3y = 6 \\ & s_2 + x + 2y = 3 \\ & x \geq 0,\ y \geq 0 \end{aligned}$$

			$-M$	$-M$	0	0	-4	-1	
BV	C	B	A_1	A_2	s_1	s_2	x	y	MR
A_1	$-M$	3	1	0	0	0	3	1	1
A_2	$-M$	6	0	1	-1	0	4	3	$\frac{3}{2}$
s_2	0	3	0	0	0	1	1	2	3
		$Z_j - C_j$	0	0	M	0	$-7M+4$	$-4M+1$	

			$-M$	0	0	-4	-1	
BV	C	B	A_2	s_1	s_2	x	y	MR
x	-4	1	0	0	0	1	$\frac{1}{3}$	3
A_2	$-M$	2	1	-1	0	0	$\frac{5}{3}$	$\frac{6}{5}$
s_2	0	2	0	0	1	0	$\frac{5}{3}$	$\frac{6}{5}$
		$Z_j - C_j$	0	M	0	0	$\frac{-5M}{3} - \frac{1}{3}$	

			0	0	-4	-1	
BV	C	B	s_1	s_2	x	y	MR
x	-4	$\frac{3}{5}$	$\frac{1}{5}$	0	1	0	3
y	-1	$\frac{6}{5}$	$-\frac{3}{5}$	0	0	1	$-$
s_2	0	0	1	1	0	0	0
		$Z_j - C_j$	$-\frac{1}{5}$	0	0	0	

			0	0	-4	-1	
BV	C	B	s_1	s_2	x	y	MR
x	-4	$\frac{3}{5}$	0	$-\frac{1}{5}$	1	0	3
y	-1	$\frac{6}{5}$	0	$\frac{3}{5}$	0	1	$-$
s_1	0	0	1	1	0	0	0
		$Z_j - C_j$	0	$\frac{1}{5}$	0	0	

$$Z = -\frac{18}{5}$$

$$x = \frac{3}{5}$$

$$y = \frac{6}{5}$$

$$\begin{aligned} \text{Maximize} \quad & 3x_1 + 2x_2 + x_3 \\ \text{subject to} \quad & -3x_1 + 2x_2 + 2x_3 = 8 \\ & -3x_1 + 4x_2 + x_3 = 7 \\ & x_1 \geq 0,\ x_2 \geq 0,\ x_3 \geq 0 \end{aligned}$$

Standard Form:

$$\begin{aligned} \text{Maximize} \quad & 3x_1 + 2x_2 + x_3 \\ \text{subject to} \quad & -3x_1 + 2x_2 + 2x_3 = 8 \\ & -3x_1 + 4x_2 + x_3 = 7 \\ & x_1 \geq 0,\ x_2 \geq 0,\ x_3 \geq 0 \end{aligned}$$

			$-M$	$-M$	3	2	1	
BV	C	B	A_1	A_2	x_1	x_2	x_3	MR
A_1	$-M$	8	1	0	-3	2	2	4
A_2	$-M$	7	0	1	-3	4	1	$\frac{7}{4}$
		$Z_j - C_j$	0	0	$6M-3$	$-6M-2$	$-3M-1$	

			$-M$	3	2	1	
BV	C	B	A_1	x_1	x_2	x_3	MR
A_1	$-M$	$\frac{9}{2}$	1	$-\frac{3}{2}$	0	$\frac{3}{2}$	3
x_2	2	$\frac{7}{4}$	0	$-\frac{3}{4}$	1	$\frac{1}{4}$	7
		$Z_j - C_j$	0	$\frac{3M}{2} - \frac{9}{2}$	0	$\frac{-3M}{2} - \frac{1}{2}$	

			3	2	1	
BV	C	B	x_1	x_2	x_3	MR
x_3	1	3	-1	0	1	$-$
x_2	2	1	$-\frac{1}{2}$	1	0	$-$
		$Z_j - C_j$	-5	0	0	

Unbounded Solution

$$x_1 + 2x_2 + x_3 = 4$$

$$2x_1 + x_2 + 5x_3 = 5$$

Determinant:

Echelon Form:

$$\begin{pmatrix} 1 & 2 \\ 0 & -3 \end{pmatrix}$$

$$\begin{pmatrix} 1 & 2 \\ 0 & -3 \end{pmatrix}$$

Inverse:

$$\left(\begin{array}{ccc|c} 1 & 2 & 1 & 0 \\ 2 & 1 & 0 & 1 \end{array}\right)$$

Echelon Form:

$$\left(\begin{array}{ccc|c} 1 & 2 & 1 & 0 \\ 0 & -3 & -2 & 1 \end{array}\right)$$

$$\left(\begin{array}{ccc|c} 1 & 2 & 1 & 0 \\ 0 & -3 & -2 & 1 \end{array}\right)$$

$$\left(\begin{array}{ccc|c} 1 & 0 & -\frac{1}{3} & \frac{2}{3} \\ 0 & 1 & \frac{2}{3} & -\frac{1}{3} \end{array}\right)$$

$$\left(\begin{array}{ccc|c} 1 & 0 & -\frac{1}{3} & \frac{2}{3} \\ 0 & 1 & \frac{2}{3} & -\frac{1}{3} \end{array}\right)$$

Determinant:
Echelon Form:

$$\begin{pmatrix} 1 & 1 \\ 0 & 3 \end{pmatrix}$$

$$\begin{pmatrix} 1 & 1 \\ 0 & 3 \end{pmatrix}$$

Inverse:

$$\left(\begin{array}{ccc|c} 1 & 1 & 1 & 0 \\ 2 & 5 & 0 & 1 \end{array}\right)$$

Echelon Form:

$$\left(\begin{array}{ccc|c} 1 & 1 & 1 & 0 \\ 0 & 3 & -2 & 1 \end{array}\right)$$

$$\left(\begin{array}{ccc|c} 1 & 1 & 1 & 0 \\ 0 & 3 & -2 & 1 \end{array}\right)$$

$$\left(\begin{array}{ccc|c} 1 & 0 & \frac{5}{3} & -\frac{1}{3} \\ 0 & 1 & -\frac{2}{3} & \frac{1}{3} \end{array}\right)$$

$$\left(\begin{array}{ccc|c} 1 & 0 & \frac{5}{3} & -\frac{1}{3} \\ 0 & 1 & -\frac{2}{3} & \frac{1}{3} \end{array}\right)$$

Determinant:
Echelon Form:

$$\begin{pmatrix} 2 & 1 \\ 0 & \frac{9}{2} \end{pmatrix}$$

$$\begin{pmatrix} 2 & 1 \\ 0 & \frac{9}{2} \end{pmatrix}$$

Inverse:

$$\left(\begin{array}{ccc|c} 2 & 1 & 1 & 0 \\ 1 & 5 & 0 & 1 \end{array}\right)$$

Echelon Form:

$$\left(\begin{array}{ccc|c} 2 & 1 & 1 & 0 \\ 0 & \frac{9}{2} & -\frac{1}{2} & 1 \end{array}\right)$$

$$\left(\begin{array}{ccc|c} 2 & 1 & 1 & 0 \\ 0 & \frac{9}{2} & -\frac{1}{2} & 1 \end{array}\right)$$

$$\left(\begin{array}{ccc|c} 2 & 0 & \frac{10}{9} & -\frac{2}{9} \\ 0 & 1 & -\frac{1}{9} & \frac{2}{9} \end{array}\right)$$

$$\left(\begin{array}{ccc|c} 1 & 0 & \frac{5}{9} & -\frac{1}{9} \\ 0 & 1 & -\frac{1}{9} & \frac{2}{9} \end{array}\right)$$

Basic Feasible Solutions:

$$x_1 = 2$$

$$x_2 = 1$$

$$x_1 = 5$$

$$x_3 = -1$$

$$x_2 = \frac{5}{3}$$

$$x_3 = \frac{2}{3}$$

$$2x_1 + x_2 - x_3 = 2$$

$$3x_1 + 2x_2 + x_3 = 3$$

Determinant:
Echelon Form:

$$\begin{pmatrix} 2 & 1 \\ 0 & \frac{1}{2} \end{pmatrix}$$

$$\begin{pmatrix} 2 & 1 \\ 0 & \frac{1}{2} \end{pmatrix}$$

Inverse:

$$\left(\begin{array}{ccc|c} 2 & 1 & 1 & 0 \\ 3 & 2 & 0 & 1 \end{array}\right)$$

Echelon Form:

$$\left(\begin{array}{ccc|c} 2 & 1 & 1 & 0 \\ 0 & \frac{1}{2} & -\frac{3}{2} & 1 \end{array}\right)$$

$$\left(\begin{array}{ccc|c} 2 & 1 & 1 & 0 \\ 0 & \frac{1}{2} & -\frac{3}{2} & 1 \end{array}\right)$$

$$\left(\begin{array}{ccc|c} 2 & 0 & 4 & -2 \\ 0 & 1 & -3 & 2 \end{array}\right)$$

$$\left(\begin{array}{ccc|c} 1 & 0 & 2 & -1 \\ 0 & 1 & -3 & 2 \end{array}\right)$$

Determinant:
Echelon Form:

$$\begin{pmatrix} 2 & -1 \\ 0 & \frac{5}{2} \end{pmatrix}$$

$$\begin{pmatrix} 2 & -1 \\ 0 & \frac{5}{2} \end{pmatrix}$$

Inverse:

$$\left(\begin{array}{ccc|c} 2 & -1 & 1 & 0 \\ 3 & 1 & 0 & 1 \end{array}\right)$$

Echelon Form:

$$\left(\begin{array}{ccc|c} 2 & -1 & 1 & 0 \\ 0 & \frac{5}{2} & -\frac{3}{2} & 1 \end{array}\right)$$

$$\left(\begin{array}{ccc|c} 2 & -1 & 1 & 0 \\ 0 & \frac{5}{2} & -\frac{3}{2} & 1 \end{array}\right)$$

$$\left(\begin{array}{ccc|c} 2 & 0 & \frac{2}{5} & \frac{2}{5} \\ 0 & 1 & -\frac{3}{5} & \frac{2}{5} \end{array}\right)$$

$$\left(\begin{array}{ccc|c} 1 & 0 & \frac{1}{5} & \frac{1}{5} \\ 0 & 1 & -\frac{3}{5} & \frac{2}{5} \end{array}\right)$$

Determinant:
Echelon Form:

$$\begin{pmatrix} 1 & -1 \\ 0 & 3 \end{pmatrix}$$

$$\begin{pmatrix} 1 & -1 \\ 0 & 3 \end{pmatrix}$$

Inverse:

$$\left(\begin{array}{ccc|c} 1 & -1 & 1 & 0 \\ 2 & 1 & 0 & 1 \end{array}\right)$$

Echelon Form:

$$\left(\begin{array}{ccc|c} 1 & -1 & 1 & 0 \\ 0 & 3 & -2 & 1 \end{array}\right)$$

$$\left(\begin{array}{ccc|c} 1 & -1 & 1 & 0 \\ 0 & 3 & -2 & 1 \end{array}\right)$$

$$\left(\begin{array}{ccc|c} 1 & 0 & \frac{1}{3} & \frac{1}{3} \\ 0 & 1 & -\frac{2}{3} & \frac{1}{3} \end{array}\right)$$

$$\left(\begin{array}{ccc|c} 1 & 0 & \frac{1}{3} & \frac{1}{3} \\ 0 & 1 & -\frac{2}{3} & \frac{1}{3} \end{array}\right)$$

Basic Feasible Solutions:

$$x_1 = 1$$

$$x_2 = 0$$

$$x_1 = 1$$

$$x_3 = 0$$

$$x_2 = \frac{5}{3}$$

$$x_3 = -\frac{1}{3}$$

$$\begin{aligned} \text{Maximize} \quad & 2x_1 + 3x_2 + 10x_3 \\ \text{subject to} \quad & x_1 + 2x_3 = 0 \\ & x_2 + x_3 = 1 \\ & x_1 \geq 0,\ x_2 \geq 0,\ x_3 \geq 0 \end{aligned}$$

Standard Form:

$$\begin{aligned} \text{Maximize} \quad & 2x_1 + 3x_2 + 10x_3 \\ \text{subject to} \quad & x_1 + 2x_3 = 0 \\ & x_2 + x_3 = 1 \\ & x_1 \geq 0,\ x_2 \geq 0,\ x_3 \geq 0 \end{aligned}$$

			2	3	10	
BV	C	B	x_1	x_2	x_3	MR
x_1	2	0	1	0	2	0
x_2	3	1	0	1	1	1
		$Z_j - C_j$	0	0	-3	

			2	3	10	
BV	C	B	x_1	x_2	x_3	MR
x_3	10	0	$\frac{1}{2}$	0	1	0
x_2	3	1	$-\frac{1}{2}$	1	0	1
		$Z_j - C_j$	$\frac{3}{2}$	0	0	

$$Z = 3$$

$$x_1 = 0$$

$$x_2 = 1$$

$$x_3 = 0$$

$$\begin{aligned} \text{Maximize} \quad & 2x_1 + 3x_2 + 10x_3 \\ \text{subject to} \quad & x_1 - 2x_3 = 0 \\ & x_2 + x_3 = 1 \\ & x_1 \geq 0,\ x_2 \geq 0,\ x_3 \geq 0 \end{aligned}$$

Standard Form:

$$\begin{aligned} \text{Maximize} \quad & 2x_1 + 3x_2 + 10x_3 \\ \text{subject to} \quad & x_1 - 2x_3 = 0 \\ & x_2 + x_3 = 1 \\ & x_1 \geq 0,\ x_2 \geq 0,\ x_3 \geq 0 \end{aligned}$$

			2	3	10	
BV	C	B	x_1	x_2	x_3	MR
x_1	2	0	1	0	-2	–
x_2	3	1	0	1	1	1
		$Z_j - C_j$	0	0	-11	

			2	3	10	
BV	C	B	x_1	x_2	x_3	MR
x_1	2	2	1	2	0	–
x_3	10	1	0	1	1	1
		$Z_j - C_j$	0	11	0	

$$Z = 14$$

$$x_1 = 2$$

$$x_2 = 0$$

$$x_3 = 1$$

$$\begin{aligned} \text{Maximize} \quad & 2x + y \\ \text{subject to} \quad & 4x + 3y \leq 12 \\ & 4x + y \leq 8 \\ & 4x - y \leq 8 \\ & x \geq 0,\ y \geq 0 \end{aligned}$$

Standard Form:

$$\begin{aligned} \text{Maximize} \quad & 2x + y \\ \text{subject to} \quad & s_1 + 4x + 3y = 12 \\ & s_2 + 4x + y = 8 \\ & s_3 + 4x - y = 8 \\ & x \geq 0,\ y \geq 0 \end{aligned}$$

			0	0	0	2	1	
BV	C	B	s_1	s_2	s_3	x	y	MR
s_1	0	12	1	0	0	4	3	3
s_2	0	8	0	1	0	4	1	2
s_3	0	8	0	0	1	4	-1	2
		$Z_j - C_j$	0	0	0	-2	-1	

			0	0	0	2	1	
BV	C	B	s_1	s_2	s_3	x	y	MR
s_1	0	4	1	0	-1	0	4	1
s_2	0	0	0	1	-1	0	2	0
x	2	2	0	0	$\frac{1}{4}$	1	$-\frac{1}{4}$	$-$
		$Z_j - C_j$	0	0	$\frac{1}{2}$	0	$-\frac{3}{2}$	

			0	0	0	2	1	
BV	C	B	s_1	s_2	s_3	x	y	MR
s_1	0	4	1	-2	1	0	0	4
y	1	0	0	$\frac{1}{2}$	$-\frac{1}{2}$	0	1	$-$
x	2	2	0	$\frac{1}{8}$	$\frac{1}{8}$	1	0	16
		$Z_j - C_j$	0	$\frac{3}{4}$	$-\frac{1}{4}$	0	0	

			0	0	0	2	1	
BV	C	B	s_1	s_2	s_3	x	y	MR
s_3	0	4	1	-2	1	0	0	4
y	1	2	$\frac{1}{2}$	$-\frac{1}{2}$	0	0	1	$-$
x	2	$\frac{3}{2}$	$-\frac{1}{8}$	$\frac{3}{8}$	0	1	0	16
		$Z_j - C_j$	$\frac{1}{4}$	$\frac{1}{4}$	0	0	0	

$$Z = 5$$

$$x = \frac{3}{2}$$

$$y = 2$$

$$\begin{aligned} \text{Maximize} \quad & 5x + 3y \\ \text{subject to} \quad & x + y \leq 2 \\ & 5x + 2y \leq 10 \\ & 3x + 8y \leq 12 \\ & x \geq 0,\ y \geq 0 \end{aligned}$$

Standard Form:

$$\begin{aligned} \text{Maximize} \quad & 5x + 3y \\ \text{subject to} \quad & s_1 + x + y = 2 \\ & s_2 + 5x + 2y = 10 \\ & s_3 + 3x + 8y = 12 \\ & x \geq 0,\ y \geq 0 \end{aligned}$$

			0	0	0	5	3	
BV	C	B	s_1	s_2	s_3	x	y	MR
s_1	0	2	1	0	0	1	1	2
s_2	0	10	0	1	0	5	2	2
s_3	0	12	0	0	1	3	8	4
		$Z_j - C_j$	0	0	0	-5	-3	

			0	0	0	5	3	
BV	C	B	s_1	s_2	s_3	x	y	MR
s_1	0	0	1	$-\frac{1}{5}$	0	0	$\frac{3}{5}$	0
x	5	2	0	$\frac{1}{5}$	0	1	$\frac{2}{5}$	5
s_3	0	6	0	$-\frac{3}{5}$	1	0	$\frac{34}{5}$	$\frac{15}{17}$
		$Z_j - C_j$	0	1	0	0	-1	

			0	0	0	5	3	
BV	C	B	s_1	s_2	s_3	x	y	MR
y	3	0	$\frac{5}{3}$	$-\frac{1}{3}$	0	0	1	0
x	5	2	$-\frac{2}{3}$	$\frac{1}{3}$	0	1	0	5
s_3	0	6	$-\frac{34}{3}$	$\frac{5}{3}$	1	0	0	$\frac{15}{17}$
		$Z_j - C_j$	$\frac{5}{3}$	$\frac{2}{3}$	0	0	0	

$$Z = 10$$

$$x = 2$$

$$y = 0$$

$$\begin{aligned} \text{Maximize} \quad & 2x + 3y \\ \text{subject to} \quad & x + 2y \geq 2 \\ & 3x + y \geq 3 \\ & 4x + 3y \leq 6 \\ & x \geq 0,\ y \geq 0 \end{aligned}$$

Standard Form:

$$\begin{aligned} \text{Maximize} \quad & 2x + 3y \\ \text{subject to} \quad & -s_1 + x + 2y = 2 \\ & -s_2 + 3x + y = 3 \\ & s_3 + 4x + 3y = 6 \\ & x \geq 0,\ y \geq 0 \end{aligned}$$

			$-M$	$-M$	0	0	0	2	3	
BV	C	B	A_1	A_2	s_1	s_2	s_3	x	y	MR
A_1	$-M$	2	1	0	-1	0	0	1	2	2
A_2	$-M$	3	0	1	0	-1	0	3	1	1
s_3	0	6	0	0	0	0	1	4	3	$\frac{3}{2}$
		$Z_j - C_j$	0	0	M	M	0	$-4M-2$	$-3M-3$	

			$-M$	0	0	0	2	3	
BV	C	B	A_1	s_1	s_2	s_3	x	y	MR
A_1	$-M$	1	1	-1	$\frac{1}{3}$	0	0	$\frac{5}{3}$	$\frac{3}{5}$
x	2	1	0	0	$-\frac{1}{3}$	0	1	$\frac{1}{3}$	3
s_3	0	2	0	0	$\frac{4}{3}$	1	0	$\frac{5}{3}$	$\frac{6}{5}$
		$Z_j - C_j$	0	M	$\frac{-M}{3} - \frac{2}{3}$	0	0	$\frac{-5M}{3} - \frac{7}{3}$	

			0	0	0	2	3	
BV	C	B	s_1	s_2	s_3	x	y	MR
y	3	$\frac{3}{5}$	$-\frac{3}{5}$	$\frac{1}{5}$	0	0	1	$-$
x	2	$\frac{4}{5}$	$\frac{1}{5}$	$-\frac{2}{5}$	0	1	0	4
s_3	0	1	1	1	1	0	0	1
		$Z_j - C_j$	$-\frac{7}{5}$	$-\frac{1}{5}$	0	0	0	

			0	0	0	2	3	
BV	C	B	s_1	s_2	s_3	x	y	MR
y	3	$\frac{6}{5}$	0	$\frac{4}{5}$	$\frac{3}{5}$	0	1	$-$
x	2	$\frac{3}{5}$	0	$-\frac{3}{5}$	$-\frac{1}{5}$	1	0	4
s_1	0	1	1	1	1	0	0	1
		$Z_j - C_j$	0	$\frac{6}{5}$	$\frac{7}{5}$	0	0	

$$Z = \frac{24}{5}$$

$$x = \frac{3}{5}$$

$$y = \frac{6}{5}$$

$$x + 2y - z = 3$$

$$3x - y + 2z = 4$$

$$2x + 3y - 5z = 7$$

Determinant:
Echelon Form:

$$\begin{pmatrix} 1 & 2 & -1 \\ 0 & -7 & 5 \\ 0 & -1 & -3 \end{pmatrix}$$

$$\begin{pmatrix} 1 & 2 & -1 \\ 0 & -7 & 5 \\ 0 & 0 & -\frac{26}{7} \end{pmatrix}$$

$$\begin{pmatrix} 1 & 2 & -1 \\ 0 & -7 & 5 \\ 0 & 0 & -\frac{26}{7} \end{pmatrix}$$

Inverse:

$$\left(\begin{array}{ccccc|c} 1 & 2 & -1 & 1 & 0 & 0 \\ 3 & -1 & 2 & 0 & 1 & 0 \\ 2 & 3 & -5 & 0 & 0 & 1 \end{array}\right)$$

Echelon Form:

$$\left(\begin{array}{ccccc|c} 1 & 2 & -1 & 1 & 0 & 0 \\ 0 & -7 & 5 & -3 & 1 & 0 \\ 0 & -1 & -3 & -2 & 0 & 1 \end{array}\right)$$

$$\left(\begin{array}{ccccc|c} 1 & 2 & -1 & 1 & 0 & 0 \\ 0 & -7 & 5 & -3 & 1 & 0 \\ 0 & 0 & -\frac{26}{7} & -\frac{11}{7} & -\frac{1}{7} & 1 \end{array}\right)$$

$$\left(\begin{array}{ccccc|c} 1 & 2 & -1 & 1 & 0 & 0 \\ 0 & -7 & 5 & -3 & 1 & 0 \\ 0 & 0 & -\frac{26}{7} & -\frac{11}{7} & -\frac{1}{7} & 1 \end{array}\right)$$

$$\left(\begin{array}{ccccc|c} 1 & 2 & 0 & \frac{37}{26} & \frac{1}{26} & -\frac{7}{26} \\ 0 & -7 & 0 & -\frac{133}{26} & \frac{21}{26} & \frac{35}{26} \\ 0 & 0 & 1 & \frac{11}{26} & \frac{1}{26} & -\frac{7}{26} \end{array}\right)$$

$$\left(\begin{array}{ccccc|c} 1 & 0 & 0 & -\frac{1}{26} & \frac{7}{26} & \frac{3}{26} \\ 0 & 1 & 0 & \frac{19}{26} & -\frac{3}{26} & -\frac{5}{26} \\ 0 & 0 & 1 & \frac{11}{26} & \frac{1}{26} & -\frac{7}{26} \end{array}\right)$$

$$\left(\begin{array}{ccccc|c} 1 & 0 & 0 & -\frac{1}{26} & \frac{7}{26} & \frac{3}{26} \\ 0 & 1 & 0 & \frac{19}{26} & -\frac{3}{26} & -\frac{5}{26} \\ 0 & 0 & 1 & \frac{11}{26} & \frac{1}{26} & -\frac{7}{26} \end{array}\right)$$

Basic Feasible Solutions:

$$x = \frac{23}{13}$$

$$y = \frac{5}{13}$$

$$z = -\frac{6}{13}$$

$$\begin{aligned} \text{Maximize} \quad & x_1 + 2x_2 + x_3 \\ \text{subject to} \quad & 2x_1 + x_2 - x_3 \geq -2 \\ & -2x_1 + x_2 - 5x_3 \leq 6 \\ & 4x_1 + x_2 + x_3 \leq 6 \\ & x_1 \geq 0,\ x_2 \geq 0,\ x_3 \geq 0 \end{aligned}$$

Standard Form:

$$\begin{aligned} \text{Maximize} \quad & x_1 + 2x_2 + x_3 \\ \text{subject to} \quad & s_1 - 2x_1 - x_2 + x_3 = 2 \\ & s_2 - 2x_1 + x_2 - 5x_3 = 6 \\ & s_3 + 4x_1 + x_2 + x_3 = 6 \\ & x_1 \geq 0,\ x_2 \geq 0,\ x_3 \geq 0 \end{aligned}$$

			0	0	0	1	2	1	
BV	C	B	s_1	s_2	s_3	x_1	x_2	x_3	MR
s_1	0	2	1	0	0	-2	-1	1	–
s_2	0	6	0	1	0	-2	1	-5	6
s_3	0	6	0	0	1	4	1	1	6
		$Z_j - C_j$	0	0	0	-1	-2	-1	

			0	0	0	1	2	1	
BV	C	B	s_1	s_2	s_3	x_1	x_2	x_3	MR
s_1	0	8	1	0	1	2	0	2	–
s_2	0	0	0	1	-1	-6	0	-6	1
x_2	2	6	0	0	1	4	1	1	0
		$Z_j - C_j$	0	0	2	7	0	1	

$$Z = 12$$

$$x_1 = 0$$

$$x_2 = 6$$

$$x_3 = 0$$

$$\begin{aligned} \text{Maximize} \quad & 2x + 3y \\ \text{subject to} \quad & x + 2y \leq 4 \\ & x + y = 3 \\ & x \geq 0,\ y \geq 0 \end{aligned}$$

Standard Form:

$$\begin{aligned} \text{Maximize} \quad & 2x + 3y \\ \text{subject to} \quad & s_1 + x + 2y = 4 \\ & x + y = 3 \\ & x \geq 0,\ y \geq 0 \end{aligned}$$

			$-M$	0	2	3	
BV	C	B	A_1	s_1	x	y	MR
s_1	0	4	0	1	1	2	4
A_1	$-M$	3	1	0	1	1	3
		$Z_j - C_j$	0	0	$-M-2$	$-M-3$	

			0	2	3	
BV	C	B	s_1	x	y	MR
s_1	0	1	1	0	1	1
x	2	3	0	1	1	3
		$Z_j - C_j$	0	0	-1	

			0	2	3	
BV	C	B	s_1	x	y	MR
y	3	1	1	0	1	1
x	2	2	-1	1	0	3
		$Z_j - C_j$	1	0	0	

$$Z = 7$$

$$x = 2$$

$$y = 1$$

$$\begin{aligned} \text{Maximize} \quad & 3x + 2y \\ \text{subject to} \quad & 4x - 3y \leq 10 \\ & x + 2y \leq 5 \\ & 3x - 5y \geq 15 \\ & x \geq 0,\ y \geq 0 \end{aligned}$$

Dual:

$$\begin{aligned} \text{Minimize} \quad & 10w_1 + 5w_2 - 15w_3 \\ \text{subject to} \quad & 4w_1 + w_2 - 3w_3 \geq 3 \\ & -3w_1 + 2w_2 + 5w_3 \geq 2 \\ & w_1 \geq 0,\ w_2 \geq 0,\ w_3 \geq 0 \end{aligned}$$

$$\begin{aligned} \text{Minimize} \quad & 5x_1 + 4x_2 - 3x_3 \\ \text{subject to} \quad & x_1 + x_2 + x_3 \geq 5 \\ & 2x_1 + 3x_2 - 5x_3 \leq 4 \\ & x_1 + 2x_2 - 3x_3 \leq 6 \\ & x_1 \geq 0,\ x_2 \geq 0,\ x_3 \geq 0 \end{aligned}$$

Dual:

$$\begin{aligned} \text{Maximize} \quad & 5w_1 - 4w_2 - 6w_3 \\ \text{subject to} \quad & w_1 - 2w_2 - w_3 \leq 5 \\ & w_1 - 3w_2 - 2w_3 \leq 4 \\ & w_1 + 5w_2 + 3w_3 \leq -3 \\ & w_1 \geq 0,\ w_2 \geq 0,\ w_3 \geq 0 \end{aligned}$$

$$\begin{aligned} \text{Maximize} \quad & x + 2y \\ \text{subject to} \quad & 3x + 2y \leq 4 \\ & 2x - 5y = 10 \\ & x \geq 0,\ y \geq 0 \end{aligned}$$

Dual:

$$\begin{aligned} \text{Minimize} \quad & 4w_1 + 10w_2 \\ \text{subject to} \quad & 3w_1 + 2w_2 \geq 1 \\ & 2w_1 - 5w_2 \geq 2 \\ & w_1 \geq 0,\ w_2 \text{ is unrestricted} \end{aligned}$$

$$\begin{aligned} \text{Minimize} \quad & 3x_1 + 4x_2 + 7x_3 \\ \text{subject to} \quad & x_1 + 2x_2 - 5x_3 \geq 4 \\ & 2x_1 + 5x_2 - x_3 = 7 \\ & 2x_1 + 3x_2 - x_3 \leq -8 \\ & x_1 \geq 0,\ x_2 \geq 0,\ x_3 \geq 0 \end{aligned}$$

Dual:

$$\begin{aligned} \text{Maximize} \quad & 4w_1 + 7w_2 + 8w_3 \\ \text{subject to} \quad & w_1 + 2w_2 - 2w_3 \leq 3 \\ & 2w_1 + 5w_2 - 3w_3 \leq 4 \\ & -5w_1 - w_2 + w_3 \leq 7 \\ & w_1 \geq 0,\ w_2 \text{ is unrestricted},\ w_3 \geq 0 \end{aligned}$$

$$\begin{aligned} \text{Maximize} \quad & 3x + 2y \\ \text{subject to} \quad & x + y \leq 5 \\ & 2x + 3y \geq 4 \\ & x - y \leq 2 \\ & x \text{ is unrestricted},\ y \geq 0 \end{aligned}$$

Dual:

$$\begin{aligned} \text{Minimize} \quad & 5w_1 - 4w_2 + 2w_3 \\ \text{subject to} \quad & w_1 - 2w_2 + w_3 = 3 \\ & w_1 - 3w_2 - w_3 \geq 2 \\ & w_1 \geq 0,\ w_2 \geq 0,\ w_3 \geq 0 \end{aligned}$$

$$\begin{aligned} \text{Maximize} \quad & x + 2y \\ \text{subject to} \quad & 2x + 3y \leq 4 \\ & 3x + 4y = 5 \\ & x \geq 0,\ y \geq 0 \end{aligned}$$

Dual:

$$\begin{aligned} \text{Minimize} \quad & 4w_1 + 5w_2 \\ \text{subject to} \quad & 2w_1 + 3w_2 \geq 1 \\ & 3w_1 + 4w_2 \geq 2 \\ & w_1 \geq 0,\ w_2 \text{ is unrestricted} \end{aligned}$$

$$\begin{aligned} \text{Maximize} \quad & -5x - 6y \\ \text{subject to} \quad & x + y \geq 2 \\ & 4x + y \geq 4 \\ & x \geq 0,\ y \geq 0 \end{aligned}$$

Canonical Form:

$$\begin{aligned} \text{Maximize} \quad & -5x - 6y \\ \text{subject to} \quad & -x - y \leq -2 \\ & -4x - y \leq -4 \\ & x \geq 0,\ y \geq 0 \end{aligned}$$

$$\begin{aligned} \text{Maximize} \quad & -5x - 6y \\ \text{subject to} \quad & -x - y \leq -2 \\ & -4x - y \leq -4 \\ & x \geq 0,\ y \geq 0 \end{aligned}$$

Standard Form:

$$\begin{aligned} \text{Maximize} \quad & -5x - 6y \\ \text{subject to} \quad & s_1 - x - y = -2 \\ & s_2 - 4x - y = -4 \\ & x \geq 0,\ y \geq 0 \end{aligned}$$

			0	0	-5	-6	
BV	C	B	s_1	s_2	x	y	MR
s_1	0	-2	1	0	-1	-1	
s_2	0	-4	0	1	-4	-1	
		$Z_j - C_j$	0	0	5	6	

			0	0	-5	-6	
BV	C	B	s_1	s_2	x	y	MR
s_1	0	-1	1	$-\frac{1}{4}$	0	$-\frac{3}{4}$	
x	-5	1	0	$-\frac{1}{4}$	1	$\frac{1}{4}$	
		$Z_j - C_j$	0	$\frac{5}{4}$	0	$\frac{19}{4}$	

			0	0	-5	-6	
BV	C	B	s_1	s_2	x	y	MR
s_2	0	4	-4	1	0	3	
x	-5	2	-1	0	1	1	
		$Z_j - C_j$	5	0	0	1	

$$Z = -10$$

$$x = 2$$

$$y = 0$$

$$\begin{aligned} \text{Minimize} \quad & 10x_1 + 6x_2 + 2x_3 \\ \text{subject to} \quad & -x_1 + x_2 + x_3 \geq 1 \\ & 3x_1 + x_2 - x_3 \geq 2 \\ & x_1 \geq 0,\ x_2 \geq 0,\ x_3 \geq 0 \end{aligned}$$

Canonical Form:

$$\begin{aligned} \text{Minimize} \quad & 10x_1 + 6x_2 + 2x_3 \\ \text{subject to} \quad & -x_1 + x_2 + x_3 \geq 1 \\ & 3x_1 + x_2 - x_3 \geq 2 \\ & x_1 \geq 0,\ x_2 \geq 0,\ x_3 \geq 0 \end{aligned}$$

$$\begin{aligned} \text{Minimize} \quad & 10x_1 + 6x_2 + 2x_3 \\ \text{subject to} \quad & -x_1 + x_2 + x_3 \geq 1 \\ & 3x_1 + x_2 - x_3 \geq 2 \\ & x_1 \geq 0,\ x_2 \geq 0,\ x_3 \geq 0 \end{aligned}$$

Standard Form:

$$\begin{aligned} \text{Maximize} \quad & -10x_1 - 6x_2 - 2x_3 \\ \text{subject to} \quad & s_1 + x_1 - x_2 - x_3 = -1 \\ & s_2 - 3x_1 - x_2 + x_3 = -2 \\ & x_1 \geq 0,\ x_2 \geq 0,\ x_3 \geq 0 \end{aligned}$$

			0	0	-10	-6	-2	
BV	C	B	s_1	s_2	x_1	x_2	x_3	MR
s_1	0	-1	1	0	1	-1	-1	
s_2	0	-2	0	1	-3	-1	1	
		$Z_j - C_j$	0	0	10	6	2	

			0	0	-10	-6	-2	
BV	C	B	s_1	s_2	x_1	x_2	x_3	MR
s_1	0	$-\frac{5}{3}$	1	$\frac{1}{3}$	0	$-\frac{4}{3}$	$-\frac{2}{3}$	
x_1	-10	$\frac{2}{3}$	0	$-\frac{1}{3}$	1	$\frac{1}{3}$	$-\frac{1}{3}$	
		$Z_j - C_j$	0	$\frac{10}{3}$	0	$\frac{8}{3}$	$\frac{16}{3}$	

			0	0	-10	-6	-2	
BV	C	B	s_1	s_2	x_1	x_2	x_3	MR
x_2	-6	$\frac{5}{4}$	$-\frac{3}{4}$	$-\frac{1}{4}$	0	1	$\frac{1}{2}$	
x_1	-10	$\frac{1}{4}$	$\frac{1}{4}$	$-\frac{1}{4}$	1	0	$-\frac{1}{2}$	
		$Z_j - C_j$	2	4	0	0	4	

$$Z = -10$$

$$x_1 = \frac{1}{4}$$

$$x_2 = \frac{5}{4}$$

$$x_3 = 0$$

$$\begin{aligned} \text{Maximize} \quad & 2x + 4y \\ \text{subject to} \quad & x + 2y \leq 5 \\ & x + y \leq 4 \\ & x \geq 0,\ y \geq 0 \end{aligned}$$

Standard Form:

$$\begin{aligned} \text{Maximize} \quad & 2x + 4y \\ \text{subject to} \quad & s_1 + x + 2y = 5 \\ & s_2 + x + y = 4 \\ & x \geq 0,\ y \geq 0 \end{aligned}$$

			0	0	2	4	
BV	C	B	s_1	s_2	x	y	MR
s_1	0	5	1	0	1	2	$\frac{5}{2}$
s_2	0	4	0	1	1	1	4
		$Z_j - C_j$	0	0	-2	-4	

			0	0	2	4	
BV	C	B	s_1	s_2	x	y	MR
y	4	$\frac{5}{2}$	$\frac{1}{2}$	0	$\frac{1}{2}$	1	$\frac{5}{2}$
s_2	0	$\frac{3}{2}$	$-\frac{1}{2}$	1	$\frac{1}{2}$	0	4
		$Z_j - C_j$	2	0	0	0	

			0	0	2	4	
BV	C	B	s_1	s_2	x	y	MR
y	4	$\frac{5}{2}$	$\frac{1}{2}$	0	$\frac{1}{2}$	1	5
s_2	0	$\frac{3}{2}$	$-\frac{1}{2}$	1	$\frac{1}{2}$	0	3
		$Z_j - C_j$	2	0	0	0	

			0	0	2	4	
BV	C	B	s_1	s_2	x	y	MR
y	4	1	1	-1	0	1	5
x	2	3	-1	2	1	0	3
		$Z_j - C_j$	2	0	0	0	

			0	0	2	4	
BV	C	B	s_1	s_2	x	y	MR
y	4	1	1	-1	0	1	$-$
x	2	3	-1	2	1	0	$\frac{3}{2}$
		$Z_j - C_j$	2	0	0	0	

			0	0	2	4	
BV	C	B	s_1	s_2	x	y	MR
y	4	$\frac{5}{2}$	$\frac{1}{2}$	0	$\frac{1}{2}$	1	$-$
s_2	0	$\frac{3}{2}$	$-\frac{1}{2}$	1	$\frac{1}{2}$	0	$\frac{3}{2}$
		$Z_j - C_j$	2	0	0	0	

$$Z = 10$$

$$x = 0$$

$$y = \frac{5}{2}$$

$$Z = 10$$

$$x = 3$$

$$y = 1$$

$$\begin{aligned} \text{Maximize} \quad & -20x - 30y \\ \text{subject to} \quad & 2x + 3y \geq 120 \\ & x + y \geq 40 \\ & 2x + \frac{3y}{2} \geq 90 \\ & x \geq 0,\ y \geq 0 \end{aligned}$$

Standard Form:

$$\begin{aligned} \text{Maximize} \quad & -20x - 30y \\ \text{subject to} \quad & -s_1 + 2x + 3y = 120 \\ & -s_2 + x + y = 40 \\ & -s_3 + 2x + \frac{3y}{2} = 90 \\ & x \geq 0,\ y \geq 0 \end{aligned}$$

			$-M$	$-M$	$-M$	0	0	0	-20	-30	
BV	C	B	A_1	A_2	A_3	s_1	s_2	s_3	x	y	MR
A_1	$-M$	120	1	0	0	-1	0	0	2	3	40
A_2	$-M$	40	0	1	0	0	-1	0	1	1	40
A_3	$-M$	90	0	0	1	0	0	-1	2	$\frac{3}{2}$	60
		$Z_j - C_j$	0	0	0	M	M	M	$-5M + 20$	$\frac{-11M}{2} + 30$	

			$-M$	$-M$	0	0	0	-20	-30	
BV	C	B	A_2	A_3	s_1	s_2	s_3	x	y	MR
y	-30	40	0	0	$-\frac{1}{3}$	0	0	$\frac{2}{3}$	1	60
A_2	$-M$	0	1	0	$\frac{1}{3}$	-1	0	$\frac{1}{3}$	0	0
A_3	$-M$	30	0	1	$\frac{1}{2}$	0	-1	1	0	30
		$Z_j - C_j$	0	0	$\frac{-5M}{6} + 10$	M	M	$\frac{-4M}{3}$	0	

			$-M$	0	0	0	-20	-30	
BV	C	B	A_3	s_1	s_2	s_3	x	y	MR
y	-30	40	0	-1	2	0	0	1	20
x	-20	0	0	1	-3	0	1	0	$-$
A_3	$-M$	30	1	$-\frac{1}{2}$	3	-1	0	0	10
		$Z_j - C_j$	0	$\frac{M}{2} + 10$	$-3M$	M	0	0	

			0	0	0	-20	-30	
BV	C	B	s_1	s_2	s_3	x	y	MR
y	-30	20	$-\frac{2}{3}$	0	$\frac{2}{3}$	0	1	20
x	-20	30	$\frac{1}{2}$	0	-1	1	0	–
s_2	0	10	$-\frac{1}{6}$	1	$-\frac{1}{3}$	0	0	10
		$Z_j - C_j$	10	0	0	0	0	

			0	0	0	-20	-30	
BV	C	B	s_1	s_2	s_3	x	y	MR
y	-30	20	$-\frac{2}{3}$	0	$\frac{2}{3}$	0	1	30
x	-20	30	$\frac{1}{2}$	0	-1	1	0	–
s_2	0	10	$-\frac{1}{6}$	1	$-\frac{1}{3}$	0	0	–
		$Z_j - C_j$	10	0	0	0	0	

			0	0	0	-20	-30	
BV	C	B	s_1	s_2	s_3	x	y	MR
s_3	0	30	-1	0	1	0	$\frac{3}{2}$	30
x	-20	60	$-\frac{1}{2}$	0	0	1	$\frac{3}{2}$	–
s_2	0	20	$-\frac{1}{2}$	1	0	0	$\frac{1}{2}$	–
		$Z_j - C_j$	10	0	0	0	0	

			0	0	0	-20	-30	
BV	C	B	s_1	s_2	s_3	x	y	MR
s_3	0	30	-1	0	1	0	$\frac{3}{2}$	20
x	-20	60	$-\frac{1}{2}$	0	0	1	$\frac{3}{2}$	40
s_2	0	20	$-\frac{1}{2}$	1	0	0	$\frac{1}{2}$	40
		$Z_j - C_j$	10	0	0	0	0	

			0	0	0	-20	-30	
BV	C	B	s_1	s_2	s_3	x	y	MR
y	-30	20	$-\frac{2}{3}$	0	$\frac{2}{3}$	0	1	20
x	-20	30	$\frac{1}{2}$	0	-1	1	0	40
s_2	0	10	$-\frac{1}{6}$	1	$-\frac{1}{3}$	0	0	40
		$Z_j - C_j$	10	0	0	0	0	

$$Z = -1200$$

$$x = 30$$

$$y = 20$$

$$Z = -1200$$

$$x = 60$$

$$y = 0$$

$$\begin{aligned} \text{Maximize} \quad & 3x + 5y \\ \text{subject to} \quad & x + y \leq 1 \\ & 2x + 3y \leq 1 \\ & x \geq 0,\ y \geq 0 \end{aligned}$$

Standard Form:

$$\begin{aligned} \text{Maximize} \quad & 3x + 5y \\ \text{subject to} \quad & s_1 + x + y = 1 \\ & s_2 + 2x + 3y = 1 \\ & x \geq 0,\ y \geq 0 \end{aligned}$$

			0	0	3	5	
BV	C	B	s_1	s_2	x	y	MR
s_1	0	1	1	0	1	1	1
s_2	0	1	0	1	2	3	$\frac{1}{3}$
		$Z_j - C_j$	0	0	-3	-5	

$$-\infty < C_x < \frac{10}{3}$$

$$\frac{9}{2} < C_y < \infty$$

$$\begin{aligned} \text{Maximize} \quad & 15x + 45y \\ \text{subject to} \quad & y \leq 50 \\ & x + 16y \leq 240 \\ & 5x + 2y \leq 162 \\ & x \geq 0,\ y \geq 0 \end{aligned}$$

Standard Form:

$$\begin{aligned} \text{Maximize} \quad & 15x + 45y \\ \text{subject to} \quad & s_1 + y = 50 \\ & s_2 + x + 16y = 240 \\ & s_3 + 5x + 2y = 162 \\ & x \geq 0,\ y \geq 0 \end{aligned}$$

			0	0	0	15	45	
BV	C	B	s_1	s_2	s_3	x	y	MR
s_1	0	50	1	0	0	0	1	50
s_2	0	240	0	1	0	1	16	15
s_3	0	162	0	0	1	5	2	81
		$Z_j - C_j$	0	0	0	-15	-45	

			0	0	0	15	45	
BV	C	B	s_1	s_2	s_3	x	y	MR
s_1	0	35	1	$-\frac{1}{16}$	0	$-\frac{1}{16}$	0	–
y	45	15	0	$\frac{1}{16}$	0	$\frac{1}{16}$	1	240
s_3	0	132	0	$-\frac{1}{8}$	1	$\frac{39}{8}$	0	$\frac{352}{13}$
		$Z_j - C_j$	0	$\frac{45}{16}$	0	$-\frac{195}{16}$	0	

$$\frac{45}{16} < C_x < \frac{225}{2}$$

$$6 < C_y < 240$$

$$\begin{aligned} \text{Maximize} \quad & x_1 + x_2 + 3x_3 \\ \text{subject to} \quad & 3x_1 + 2x_2 + x_3 \leq 3 \\ & 2x_1 + x_2 + 2x_3 \leq 2 \\ & x_1 \geq 0,\ x_2 \geq 0,\ x_3 \geq 0 \end{aligned}$$

Standard Form:

$$\begin{aligned} \text{Maximize} \quad & x_1 + x_2 + 3x_3 \\ \text{subject to} \quad & s_1 + 3x_1 + 2x_2 + x_3 = 3 \\ & s_2 + 2x_1 + x_2 + 2x_3 = 2 \\ & x_1 \geq 0,\ x_2 \geq 0,\ x_3 \geq 0 \end{aligned}$$

			0	0	1	1	3	
BV	C	B	s_1	s_2	x_1	x_2	x_3	MR
s_1	0	3	1	0	3	2	1	3
s_2	0	2	0	1	2	1	2	1
		$Z_j - C_j$	0	0	-1	-1	-3	

$$-\infty < C_{x1} < 3$$

$$-\infty < C_{x2} < \frac{3}{2}$$

$$2 < C_{x3} < \infty$$

$$\begin{aligned}
\text{Maximize} \quad & -x_1 + 2x_2 - x_3 \\
\text{subject to} \quad & 3x_1 + x_2 - x_3 \leq 10 \\
& -x_1 + 4x_2 + x_3 \geq 6 \\
& x_2 + x_3 \leq 4 \\
& x_1 \geq 0,\ x_2 \geq 0,\ x_3 \geq 0
\end{aligned}$$

Standard Form:

$$\begin{aligned}
\text{Maximize} \quad & -x_1 + 2x_2 - x_3 \\
\text{subject to} \quad & s_1 + 3x_1 + x_2 - x_3 = 10 \\
& -s_2 - x_1 + 4x_2 + x_3 = 6 \\
& s_3 + x_2 + x_3 = 4 \\
& x_1 \geq 0,\ x_2 \geq 0,\ x_3 \geq 0
\end{aligned}$$

			$-M$	0	0	0	-1	2	-1	
BV	C	B	A_1	s_1	s_2	s_3	x_1	x_2	x_3	MR
s_1	0	10	0	1	0	0	3	1	-1	10
A_1	$-M$	6	1	0	-1	0	-1	4	1	$\frac{3}{2}$
s_3	0	4	0	0	0	1	0	1	1	4
		$Z_j - C_j$	0	0	M	0	$M+1$	$-4M-2$	$-M+1$	

			0	0	0	-1	2	-1	
BV	C	B	s_1	s_2	s_3	x_1	x_2	x_3	MR
s_1	0	$\frac{17}{2}$	1	$\frac{1}{4}$	0	$\frac{13}{4}$	0	$-\frac{5}{4}$	34
x_2	2	$\frac{3}{2}$	0	$-\frac{1}{4}$	0	$-\frac{1}{4}$	1	$\frac{1}{4}$	–
s_3	0	$\frac{5}{2}$	0	$\frac{1}{4}$	1	$\frac{1}{4}$	0	$\frac{3}{4}$	10
		$Z_j - C_j$	0	$-\frac{1}{2}$	0	$\frac{1}{2}$	0	$\frac{3}{2}$	

$$4 < B_1 < \infty$$

$$-4 < B_2 < \infty$$

$$\frac{3}{2} < B_3 < 10$$

$$\begin{aligned} \text{Maximize} \quad & 2x + y \\ \text{subject to} \quad & 3x + 5y \leq 15 \\ & 6x + 2y \leq 24 \\ & x \geq 0,\ y \geq 0 \end{aligned}$$

Standard Form:

$$\begin{aligned} \text{Maximize} \quad & 2x + y \\ \text{subject to} \quad & s_1 + 3x + 5y = 15 \\ & s_2 + 6x + 2y = 24 \\ & x \geq 0,\ y \geq 0 \end{aligned}$$

			0	0	2	1	
BV	C	B	s_1	s_2	x	y	MR
s_1	0	15	1	0	3	5	5
s_2	0	24	0	1	6	2	4
		$Z_j - C_j$	0	0	-2	-1	

			0	0	2	1	
BV	C	B	s_1	s_2	x	y	MR
s_1	0	3	1	$-\frac{1}{2}$	0	4	$\frac{3}{4}$
x	2	4	0	$\frac{1}{6}$	1	$\frac{1}{3}$	12
		$Z_j - C_j$	0	$\frac{1}{3}$	0	$-\frac{1}{3}$	

$$12 < B_1 < 60$$

$$6 < B_2 < 30$$

$$\begin{aligned} \text{Maximize} \quad & 15x + 10y \\ \text{subject to} \quad & 4x + 6y \leq 360 \\ & 3x \leq 180 \\ & 5y \leq 200 \\ & x \geq 0,\ y \geq 0 \end{aligned}$$

Standard Form:

$$\begin{aligned} \text{Maximize} \quad & 15x + 10y \\ \text{subject to} \quad & s_1 + 4x + 6y = 360 \\ & s_2 + 3x = 180 \\ & s_3 + 5y = 200 \\ & x \geq 0,\ y \geq 0 \end{aligned}$$

			0	0	0	15	10	
BV	C	B	s_1	s_2	s_3	x	y	MR
s_1	0	360	1	0	0	4	6	90
s_2	0	180	0	1	0	3	0	60
s_3	0	200	0	0	1	0	5	–
		$Z_j - C_j$	0	0	0	−15	−10	

			0	0	0	15	10	
BV	C	B	s_1	s_2	s_3	x	y	MR
s_1	0	120	1	$-\frac{4}{3}$	0	0	6	20
x	15	60	0	$\frac{1}{3}$	0	1	0	–
s_3	0	200	0	0	1	0	5	40
		$Z_j - C_j$	0	5	0	0	−10	

$$240 < B_1 < 480$$

$$90 < B_2 < 270$$

$$100 < B_3 < \infty$$

			0	0	2	3	1	
BV	C	B	s_1	s_2	x	y	z	MR
x	2	1	4	-1	1	0	-1	
y	3	2	-1	1	0	1	2	
		$Z_j - C_j$	5	1	0	0	3	

$$\frac{11}{4} < B_1 < 5$$

$$5 < B_2 < 8$$

$$\begin{aligned} \text{Maximize} \quad & 3x + 5y \\ \text{subject to} \quad & x \leq 4 \\ & 3x + 2y \leq 18 \\ & x \geq 0,\ y \geq 0 \end{aligned}$$

Standard Form:

$$\begin{aligned} \text{Maximize} \quad & 3x + 5y \\ \text{subject to} \quad & s_1 + x = 4 \\ & s_2 + 3x + 2y = 18 \\ & x \geq 0,\ y \geq 0 \end{aligned}$$

			0	0	3	5	
BV	C	B	s_1	s_2	x	y	MR
s_1	0	4	1	0	1	0	–
s_2	0	18	0	1	3	2	9
		$Z_j - C_j$	0	0	-3	-5	

			0	0	3	5	
BV	C	B	s_1	s_2	x	y	MR
s_1	0	4	1	0	1	0	–
y	5	9	0	$\frac{1}{2}$	$\frac{3}{2}$	1	9
		$Z_j - C_j$	0	$\frac{5}{2}$	$\frac{9}{2}$	0	

			0	0	3	7	5	
BV	C	B	s_1	s_2	x	x_1	y	MR
s_1	0	4	1	0	1	0	0	–
y	5	9	0	$\frac{1}{2}$	$\frac{3}{2}$	0	1	9
		$Z_j - C_j$	0	$\frac{5}{2}$	$\frac{9}{2}$	0		

			0	0	3	7	5	
BV	C	B	s_1	s_2	x	x_1	y	MR
s_1	0	4	1	0	1	0	0	–
y	5	9	0	$\frac{1}{2}$	$\frac{3}{2}$	0	1	–
		$Z_j - C_j$	0	$\frac{5}{2}$	$\frac{9}{2}$	-7	0	

Unbounded Solution

$$\begin{aligned} \text{Maximize} \quad & 3x + 5y \\ \text{subject to} \quad & x \leq 4 \\ & 3x + 2y \leq 18 \\ & x \geq 0,\ y \geq 0 \end{aligned}$$

Standard Form:

$$\begin{aligned} \text{Maximize} \quad & 3x + 5y \\ \text{subject to} \quad & s_1 + x = 4 \\ & s_2 + 3x + 2y = 18 \\ & x \geq 0,\ y \geq 0 \end{aligned}$$

			0	0	3	5	
BV	C	B	s_1	s_2	x	y	MR
s_1	0	4	1	0	1	0	–
s_2	0	18	0	1	3	2	9
		$Z_j - C_j$	0	0	-3	-5	

			0	0	3	5	
BV	C	B	s_1	s_2	x	y	MR
s_1	0	4	1	0	1	0	–
y	5	9	0	$\frac{1}{2}$	$\frac{3}{2}$	1	9
		$Z_j - C_j$	0	$\frac{5}{2}$	$\frac{9}{2}$	0	

			0	0	5	
BV	C	B	s_1	s_2	y	MR
s_1	0	4	1	0	0	–
y	5	9	0	$\frac{1}{2}$	1	9
		$Z_j - C_j$	0	$\frac{5}{2}$	0	

			0	0	5	
BV	C	B	s_1	s_2	y	MR
s_1	0	4	1	0	0	–
y	5	9	0	$\frac{1}{2}$	1	9
		$Z_j - C_j$	0	$\frac{5}{2}$	0	

$$Z = 45$$

$$y = 9$$

$$\begin{aligned} \text{Maximize} \quad & 3x + 5y \\ \text{subject to} \quad & x \leq 4 \\ & 3x + 2y \leq 18 \\ & x \geq 0,\ y \geq 0 \end{aligned}$$

Standard Form:

$$\begin{aligned} \text{Maximize} \quad & 3x + 5y \\ \text{subject to} \quad & s_1 + x = 4 \\ & s_2 + 3x + 2y = 18 \\ & x \geq 0,\ y \geq 0 \end{aligned}$$

			0	0	3	5	
BV	C	B	s_1	s_2	x	y	MR
s_1	0	4	1	0	1	0	–
s_2	0	18	0	1	3	2	9
		$Z_j - C_j$	0	0	-3	-5	

			0	0	3	5	
BV	C	B	s_1	s_2	x	y	MR
s_1	0	4	1	0	1	0	–
y	5	9	0	$\frac{1}{2}$	$\frac{3}{2}$	1	9
		$Z_j - C_j$	0	$\frac{5}{2}$	$\frac{9}{2}$	0	

			0	0	3	$-M$	
BV	C	B	s_1	s_2	x	y	MR
s_1	0	4	1	0	1	0	–
y	$-M$	9	0	$\frac{1}{2}$	$\frac{3}{2}$	1	9
		$Z_j - C_j$	0	$\frac{5}{2}$	$\frac{9}{2}$	0	

			0	0	3	$-M$	
BV	C	B	s_1	s_2	x	y	MR
s_1	0	4	1	0	1	0	4
y	$-M$	9	0	$\frac{1}{2}$	$\frac{3}{2}$	1	6
		$Z_j - C_j$	0	$\frac{-M}{2}$	$\frac{-3M}{2} - 3$	0	

			0	0	3	$-M$	
BV	C	B	s_1	s_2	x	y	MR
x	3	4	1	0	1	0	$-$
y	$-M$	3	$-\frac{3}{2}$	$\frac{1}{2}$	0	1	6
		$Z_j - C_j$	$\frac{3M}{2} + 3$	$\frac{-M}{2}$	0	0	

			0	0	3	
BV	C	B	s_1	s_2	x	MR
x	3	4	1	0	1	$-$
s_2	0	6	-3	1	0	6
		$Z_j - C_j$	3	0	0	

$$Z = 12$$

$$x = 4$$

			0	0	0	2	4	1	3	2	
BV	C	B	s_1	s_2	s_3	x_1	x_2	x_3	x_4	x_5	MR
x_1	2	3	$\frac{1}{2}$	$\frac{1}{5}$	-1	1	0	0	-1	0	
x_2	4	1	-1	0	$\frac{1}{2}$	0	1	0	2	1	
x_3	1	7	5	$-\frac{3}{10}$	2	0	0	1	-1	-2	
		$Z_j - C_j$	2	$\frac{1}{10}$	2	0	0	0	2	0	

			0	0	0	2	4	1	3	2	
BV	C	B	s_1	s_2	s_3	x_1	x_2	x_3	x_4	x_5	MR
x_1	2	3	$\frac{1}{2}$	$\frac{1}{5}$	-1	1	0	0	-1	0	
x_2	4	1	-1	0	$\frac{1}{2}$	0	1	0	2	1	
x_3	1	7	5	$-\frac{3}{10}$	2	0	0	1	-1	-2	
		$Z_j - C_j$	2	$\frac{1}{10}$	2	0	0	0	2	0	

			0	0	0	2	$-M$	1	3	2	
BV	C	B	s_1	s_2	s_3	x_1	x_2	x_3	x_4	x_5	MR
x_1	2	3	$\frac{1}{2}$	$\frac{1}{5}$	-1	1	0	0	-1	0	
x_2	$-M$	1	-1	0	$\frac{1}{2}$	0	1	0	2	1	
x_3	1	7	5	$-\frac{3}{10}$	2	0	0	1	-1	-2	
		$Z_j - C_j$	2	$\frac{1}{10}$	2	0	0	0	2	0	

			0	0	0	2	$-M$	1	3	2	
BV	C	B	s_1	s_2	s_3	x_1	x_2	x_3	x_4	x_5	MR
x_1	2	3	$\frac{1}{2}$	$\frac{1}{5}$	-1	1	0	0	-1	0	$-$
x_2	$-M$	1	-1	0	$\frac{1}{2}$	0	1	0	2	1	$\frac{1}{2}$
x_3	1	7	5	$-\frac{3}{10}$	2	0	0	1	-1	-2	$-$
		$Z_j - C_j$	$M+6$	$\frac{1}{10}$	$\frac{-M}{2}$	0	0	0	$-2M-6$	$-M-4$	

			0	0	0	2	1	3	2	
BV	C	B	s_1	s_2	s_3	x_1	x_3	x_4	x_5	MR
x_1	2	$\frac{7}{2}$	0	$\frac{1}{5}$	$-\frac{3}{4}$	1	0	0	$\frac{1}{2}$	7
x_4	3	$\frac{1}{2}$	$-\frac{1}{2}$	0	$\frac{1}{4}$	0	0	1	$\frac{1}{2}$	1
x_3	1	$\frac{15}{2}$	$\frac{9}{2}$	$-\frac{3}{10}$	$\frac{9}{4}$	0	1	0	$-\frac{3}{2}$	$-$
		$Z_j - C_j$	3	$\frac{1}{10}$	$\frac{3}{2}$	0	0	0	-1	

			0	0	0	2	1	3	2	
BV	C	B	s_1	s_2	s_3	x_1	x_3	x_4	x_5	MR
x_1	2	3	$\frac{1}{2}$	$\frac{1}{5}$	-1	1	0	-1	0	7
x_5	2	1	-1	0	$\frac{1}{2}$	0	0	2	1	1
x_3	1	9	3	$-\frac{3}{10}$	3	0	1	3	0	$-$
		$Z_j - C_j$	2	$\frac{1}{10}$	2	0	0	2	0	

$$Z = -17$$

$$x_1 = 3$$

$$x_3 = 9$$

$$x_4 = 0$$

$$x_5 = 1$$

			0	0	0	2	4	1	3	2	
BV	C	B	s_1	s_2	s_3	x_1	x_2	x_3	x_4	x_5	MR
x_1	2	3	$\frac{1}{2}$	$-\frac{1}{5}$	-1	1	0	0	-1	0	
x_2	4	1	-1	0	$\frac{1}{2}$	0	1	0	2	1	
x_3	1	7	5	$-\frac{2}{5}$	2	0	0	1	-1	-2	
		$Z_j - C_j$	2	$-\frac{4}{5}$	2	0	0	0	2	0	

			0	0	0	2	4	1	3	2	
BV	C	B	s_1	s_2	s_3	x_1	x_2	x_3	x_4	x_5	MR
x_1	2	3	$\frac{1}{2}$	$-\frac{1}{5}$	-1	1	0	0	-1	0	
x_2	4	1	-1	0	$\frac{1}{2}$	0	1	0	2	1	
x_3	1	7	5	$-\frac{2}{5}$	2	0	0	1	-1	-2	
		$Z_j - C_j$	2	$-\frac{4}{5}$	2	0	0	0	2	0	

$$Z = -17$$

$$x_1 = 3$$

$$x_2 = 1$$

$$x_3 = 7$$

$$x_4 = 0$$

$$x_5 = 0$$

			0	0	0	2	4	1	3	2	
BV	C	B	s_1	s_2	s_3	x_1	x_2	x_3	x_4	x_5	MR
x_1	2	3	$\frac{1}{2}$	$-\frac{1}{5}$	-1	1	0	0	-1	0	
x_2	4	1	-1	0	$\frac{1}{2}$	0	1	0	2	1	
x_3	1	7	5	$-\frac{2}{5}$	2	0	0	1	-1	-2	
		$Z_j - C_j$	2	$-\frac{4}{5}$	2	0	0	0	2	0	

			0	0	0	2	4	1	3	2	
BV	C	B	s_1	s_2	s_3	x_1	x_2	x_3	x_4	x_5	MR
x_1	2	3	$\frac{1}{2}$	$-\frac{1}{5}$	-1	1	0	0	-1	0	
x_2	4	1	-1	0	$\frac{1}{2}$	0	1	0	2	1	
x_3	1	7	5	$-\frac{2}{5}$	2	0	0	1	-1	-2	
		$Z_j - C_j$	2	$-\frac{4}{5}$	2	0	0	0	2	0	

$$Z = -17$$

$$x_1 = 3$$

$$x_2 = 1$$

$$x_3 = 7$$

$$x_4 = 0$$

$$x_5 = 0$$

			0	0	0	2	4	1	3	2	
BV	C	B	s_1	s_2	s_3	x_1	x_2	x_3	x_4	x_5	MR
x_1	2	3	$\frac{1}{2}$	$-\frac{1}{5}$	-1	1	0	0	-1	0	
x_2	4	1	-1	0	$\frac{1}{2}$	0	1	0	2	1	
x_3	1	7	5	$-\frac{2}{5}$	2	0	0	1	-1	-2	
		$Z_j - C_j$	2	$-\frac{4}{5}$	2	0	0	0	2	0	

			0	0	0	2	4	1	3	2	
BV	C	B	s_1	s_2	s_3	x_1	x_2	x_3	x_4	x_5	MR
x_1	2	3	$\frac{1}{2}$	$-\frac{1}{5}$	-1	1	0	0	-1	0	
x_2	4	1	-1	0	$\frac{1}{2}$	0	1	0	2	1	
x_3	1	7	5	$-\frac{2}{5}$	2	0	0	1	-1	-2	
		$Z_j - C_j$	2	$-\frac{4}{5}$	2	0	0	0	2	0	

$$Z = -17$$

$$x_1 = 3$$

$$x_2 = 1$$

$$x_3 = 7$$

$$x_4 = 0$$

$$x_5 = 0$$

			0	0	0	2	4	1	3	2	
BV	C	B	s_1	s_2	s_3	x_1	x_2	x_3	x_4	x_5	MR
x_1	2	3	$\frac{1}{2}$	$-\frac{1}{5}$	-1	1	0	0	-1	0	
x_2	4	1	-1	0	$\frac{1}{2}$	0	1	0	2	1	
x_3	1	7	5	$-\frac{2}{5}$	2	0	0	1	-1	-2	
		$Z_j - C_j$	2	$-\frac{4}{5}$	2	0	0	0	2	0	

			0	0	0	0	2	4	1	3	2	
BV	C	B	s_1	s_2	s_3	s_4	x_1	x_2	x_3	x_4	x_5	MR
x_1	2	3	$\frac{1}{2}$	$-\frac{1}{5}$	-1	0	1	0	0	-1	0	
x_2	4	1	-1	0	$\frac{1}{2}$	0	0	1	0	2	1	
x_3	1	7	5	$-\frac{2}{5}$	2	0	0	0	1	-1	-2	
s_4	0	-5	$-\frac{21}{2}$	$\frac{7}{5}$	$-\frac{3}{2}$	1	0	0	0	4	12	
		$Z_j - C_j$	2	$-\frac{4}{5}$	2	0	0	0	2	0		

			0	0	0	0	2	4	1	3	2	
BV	C	B	s_1	s_2	s_3	s_4	x_1	x_2	x_3	x_4	x_5	MR
x_1	2	3	$\frac{1}{2}$	$-\frac{1}{5}$	-1	0	1	0	0	-1	0	$-$
x_2	4	1	-1	0	$\frac{1}{2}$	0	0	1	0	2	1	$-$
x_3	1	7	5	$-\frac{2}{5}$	2	0	0	0	1	-1	-2	$-$
s_4	0	-5	$-\frac{21}{2}$	$\frac{7}{5}$	$-\frac{3}{2}$	1	0	0	0	4	12	$-\frac{25}{7}$
		$Z_j - C_j$	2	$-\frac{4}{5}$	2	0	0	0	0	2	0	

			0	0	0	0	2	4	1	3	2	
BV	C	B	s_1	s_2	s_3	s_4	x_1	x_2	x_3	x_4	x_5	MR
x_1	2	$\frac{16}{7}$	-1	0	$-\frac{17}{14}$	$\frac{1}{7}$	1	0	0	$-\frac{3}{7}$	$\frac{12}{7}$	–
x_2	4	1	-1	0	$\frac{1}{2}$	0	0	1	0	2	1	–
x_3	1	$\frac{39}{7}$	2	0	$\frac{11}{7}$	$\frac{2}{7}$	0	0	1	$\frac{1}{7}$	$\frac{10}{7}$	$\frac{39}{14}$
s_2	0	$-\frac{25}{7}$	$-\frac{15}{2}$	1	$-\frac{15}{14}$	$\frac{5}{7}$	0	0	0	$\frac{20}{7}$	$\frac{60}{7}$	–
		$Z_j - C_j$	-4	0	$\frac{8}{7}$	$\frac{4}{7}$	0	0	0	$\frac{30}{7}$	$\frac{48}{7}$	

			0	0	0	0	2	4	1	3	2	
BV	C	B	s_1	s_2	s_3	s_4	x_1	x_2	x_3	x_4	x_5	MR
x_1	2	$\frac{71}{14}$	0	0	$-\frac{3}{7}$	$\frac{2}{7}$	1	0	$\frac{1}{2}$	$-\frac{5}{14}$	$\frac{17}{7}$	–
x_2	4	$\frac{53}{14}$	0	0	$\frac{9}{7}$	$\frac{1}{7}$	0	1	$\frac{1}{2}$	$\frac{29}{14}$	$\frac{12}{7}$	–
s_1	0	$\frac{39}{14}$	1	0	$\frac{11}{14}$	$\frac{1}{7}$	0	0	$\frac{1}{2}$	$\frac{1}{14}$	$\frac{5}{7}$	$\frac{39}{14}$
s_2	0	$\frac{485}{28}$	0	1	$\frac{135}{28}$	$\frac{25}{14}$	0	0	$\frac{15}{4}$	$\frac{95}{28}$	$\frac{195}{14}$	–
		$Z_j - C_j$	0	0	$\frac{30}{7}$	$\frac{8}{7}$	0	0	2	$\frac{32}{7}$	$\frac{68}{7}$	

$$Z = -\frac{177}{7}$$

$$x_1 = \frac{71}{14}$$

$$x_2 = \frac{53}{14}$$

$$x_3 = 0$$

$$x_4 = 0$$

$$x_5 = 0$$

$$\begin{aligned} \text{Maximize} \quad & 3x - 5y \\ \text{subject to} \quad & 4x + 3y \geq 5 \\ & 2x - 5y \leq 3 \\ & x \geq 0,\ y \geq 0 \end{aligned}$$

Dual:

$$\begin{aligned} \text{Minimize} \quad & -5w_1 + 3w_2 \\ \text{subject to} \quad & -4w_1 + 2w_2 \geq 3 \\ & -3w_1 - 5w_2 \geq -5 \\ & w_1 \geq 0,\ w_2 \geq 0 \end{aligned}$$

$$\begin{aligned} \text{Maximize} \quad & 2x + y \\ \text{subject to} \quad & x - y \leq 10 \\ & 2x - y \leq 40 \\ & x \geq 0,\ y \geq 0 \end{aligned}$$

Standard Form:

$$\begin{aligned} \text{Maximize} \quad & 2x + y \\ \text{subject to} \quad & s_1 + x - y = 10 \\ & s_2 + 2x - y = 40 \\ & x \geq 0,\ y \geq 0 \end{aligned}$$

			0	0	2	1	
BV	C	B	s_1	s_2	x	y	MR
s_1	0	10	1	0	1	-1	10
s_2	0	40	0	1	2	-1	20
		$Z_j - C_j$	0	0	-2	-1	

			0	0	2	1	
BV	C	B	s_1	s_2	x	y	MR
x	2	10	1	0	1	-1	–
s_2	0	20	-2	1	0	1	20
		$Z_j - C_j$	2	0	0	-3	

			0	0	2	1	
BV	C	B	s_1	s_2	x	y	MR
x	2	30	-1	1	1	0	$-$
y	1	20	-2	1	0	1	$-$
		$Z_j - C_j$	-4	3	0	0	

Unbounded Solution

$$\begin{aligned} \text{Maximize} \quad & 5x - y \\ \text{subject to} \quad & 3x + 5y \leq 15 \\ & 4x + 3y \leq 12 \\ & x \geq 0,\ y \geq 0 \end{aligned}$$

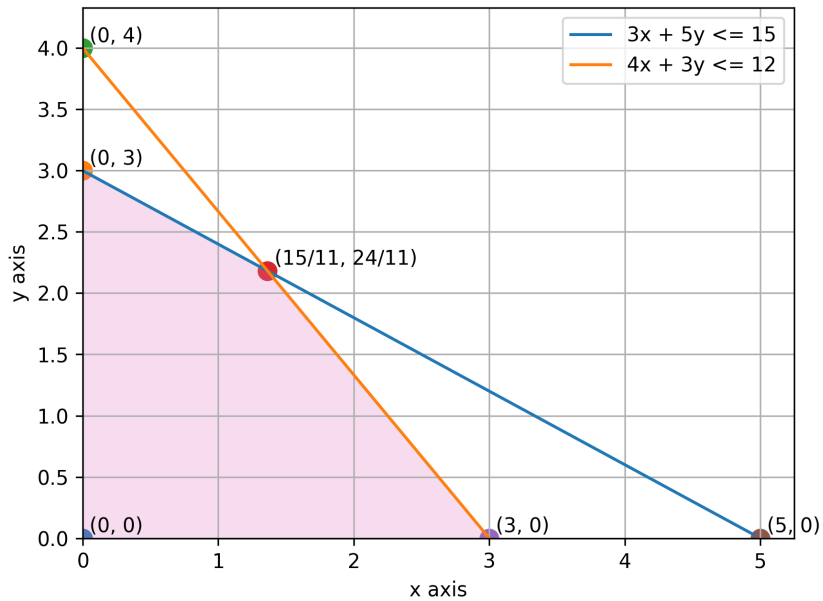
Gauss Elimination:
Echelon Form:

$$\left(\begin{array}{cc|c} 3 & 5 & 15 \\ 0 & -\frac{11}{3} & -8 \end{array}\right)$$

$$\left(\begin{array}{cc|c} 3 & 5 & 15 \\ 0 & -\frac{11}{3} & -8 \end{array}\right)$$

$$x = \frac{15}{11}$$

$$y = \frac{24}{11}$$



$$Z = 15$$

$$x = 3$$

$$y = 0$$

$$\begin{aligned} \text{Maximize} \quad & 5x - y \\ \text{subject to} \quad & 3x + 5y \leq 15 \\ & 4x + 3y \leq 12 \\ & x \geq 0,\ y \geq 0 \end{aligned}$$

Standard Form:

$$\begin{aligned} \text{Maximize} \quad & 5x - y \\ \text{subject to} \quad & s_1 + 3x + 5y = 15 \\ & s_2 + 4x + 3y = 12 \\ & x \geq 0,\ y \geq 0 \end{aligned}$$

			0	0	5	−1	
BV	C	B	s_1	s_2	x	y	MR
s_1	0	15	1	0	3	5	5
s_2	0	12	0	1	4	3	3
		$Z_j - C_j$	0	0	−5	1	

			0	0	5	−1	
BV	C	B	s_1	s_2	x	y	MR
s_1	0	6	1	$-\frac{3}{4}$	0	$\frac{11}{4}$	5
x	5	3	0	$\frac{1}{4}$	1	$\frac{3}{4}$	3
		$Z_j - C_j$	0	$\frac{5}{4}$	0	$\frac{19}{4}$	

$$Z = 15$$

$$x = 3$$

$$y = 0$$

$$\begin{aligned} \text{Maximize} \quad & x_1 + x_2 \\ \text{subject to} \quad & x_1 + x_2 \leq 8 \\ & 2x_1 + x_2 \leq 10 \\ & x_1 \geq 0,\ x_2 \geq 0 \end{aligned}$$

Standard Form:

$$\begin{aligned} \text{Maximize} \quad & x_1 + x_2 \\ \text{subject to} \quad & s_1 + x_1 + x_2 = 8 \\ & s_2 + 2x_1 + x_2 = 10 \\ & x_1 \geq 0,\ x_2 \geq 0 \end{aligned}$$

			0	0	1	1	
BV	C	B	s_1	s_2	x_1	x_2	MR
s_1	0	8	1	0	1	1	8
s_2	0	10	0	1	2	1	5
		$Z_j - C_j$	0	0	−1	−1	

			0	0	1	1	
BV	C	B	s_1	s_2	x_1	x_2	MR
s_1	0	3	1	$-\frac{1}{2}$	0	$\frac{1}{2}$	6
x_1	1	5	0	$\frac{1}{2}$	1	$\frac{1}{2}$	10
		$Z_j - C_j$	0	$\frac{1}{2}$	0	$-\frac{1}{2}$	

$$5 < B_1 < 10$$

$$8 < B_2 < 16$$

$$\begin{aligned} \text{Maximize} \quad & -4x - y \\ \text{subject to} \quad & 3x + y = 3 \\ & 4x + 3y \geq 6 \\ & x + 2y \leq 3 \\ & x \geq 0,\ y \geq 0 \end{aligned}$$

Standard Form:

$$\begin{aligned} \text{Maximize} \quad & -4x - y \\ \text{subject to} \quad & 3x + y = 3 \\ & -s_1 + 4x + 3y = 6 \\ & s_2 + x + 2y = 3 \\ & x \geq 0,\ y \geq 0 \end{aligned}$$

			$-M$	$-M$	0	0	-4	-1	
BV	C	B	A_1	A_2	s_1	s_2	x	y	MR
A_1	$-M$	3	1	0	0	0	3	1	1
A_2	$-M$	6	0	1	-1	0	4	3	$\frac{3}{2}$
s_2	0	3	0	0	0	1	1	2	3
		$Z_j - C_j$	0	0	M	0	$-7M + 4$	$-4M + 1$	

			$-M$	0	0	-4	-1	
BV	C	B	A_2	s_1	s_2	x	y	MR
x	-4	1	0	0	0	1	$\frac{1}{3}$	3
A_2	$-M$	2	1	-1	0	0	$\frac{5}{3}$	$\frac{6}{5}$
s_2	0	2	0	0	1	0	$\frac{5}{3}$	$\frac{6}{5}$
		$Z_j - C_j$	0	M	0	0	$\frac{-5M}{3} - \frac{1}{3}$	

			0	0	-4	-1	
BV	C	B	s_1	s_2	x	y	MR
x	-4	$\frac{3}{5}$	$\frac{1}{5}$	0	1	0	3
y	-1	$\frac{6}{5}$	$-\frac{3}{5}$	0	0	1	$-$
s_2	0	0	1	1	0	0	0
		$Z_j - C_j$	$-\frac{1}{5}$	0	0	0	

			0	0	-4	-1	
BV	C	B	s_1	s_2	x	y	MR
x	-4	$\frac{3}{5}$	0	$-\frac{1}{5}$	1	0	3
y	-1	$\frac{6}{5}$	0	$\frac{3}{5}$	0	1	$-$
s_1	0	0	1	1	0	0	0
		$Z_j - C_j$	0	$\frac{1}{5}$	0	0	

$$Z = -\frac{18}{5}$$

$$x = \frac{3}{5}$$

$$y = \frac{6}{5}$$

$$\begin{aligned} \text{Minimize} \quad & x_1 - 3x_2 + 2x_3 \\ \text{subject to} \quad & 3x_1 - x_2 + 2x_3 \leq 7 \\ & -2x_1 + 4x_2 \leq 12 \\ & -4x_1 + 3x_2 + 8x_3 \leq 10 \\ & x_1 \geq 0,\ x_2 \geq 0,\ x_3 \geq 0 \end{aligned}$$

Standard Form:

$$\begin{aligned} \text{Maximize} \quad & -x_1 + 3x_2 - 2x_3 \\ \text{subject to} \quad & s_1 + 3x_1 - x_2 + 2x_3 = 7 \\ & s_2 - 2x_1 + 4x_2 = 12 \\ & s_3 - 4x_1 + 3x_2 + 8x_3 = 10 \\ & x_1 \geq 0,\ x_2 \geq 0,\ x_3 \geq 0 \end{aligned}$$

			0	0	0	−1	3	−2	
BV	C	B	s_1	s_2	s_3	x_1	x_2	x_3	MR
s_1	0	7	1	0	0	3	−1	2	–
s_2	0	12	0	1	0	−2	4	0	3
s_3	0	10	0	0	1	−4	3	8	$\frac{10}{3}$
		$Z_j - C_j$	0	0	0	1	−3	2	

			0	0	0	−1	3	−2	
BV	C	B	s_1	s_2	s_3	x_1	x_2	x_3	MR
s_1	0	10	1	$\frac{1}{4}$	0	$\frac{5}{2}$	0	2	4
x_2	3	3	0	$\frac{1}{4}$	0	$-\frac{1}{2}$	1	0	–
s_3	0	1	0	$-\frac{3}{4}$	1	$-\frac{5}{2}$	0	8	–
		$Z_j - C_j$	0	$\frac{3}{4}$	0	$-\frac{1}{2}$	0	2	

			0	0	0	−1	3	−2	
BV	C	B	s_1	s_2	s_3	x_1	x_2	x_3	MR
x_1	−1	4	$\frac{2}{5}$	$\frac{1}{10}$	0	1	0	$\frac{4}{5}$	4
x_2	3	5	$\frac{1}{5}$	$\frac{3}{10}$	0	0	1	$\frac{2}{5}$	–
s_3	0	11	1	$-\frac{1}{2}$	1	0	0	10	–
		$Z_j - C_j$	$\frac{1}{5}$	$\frac{4}{5}$	0	0	0	$\frac{12}{5}$	

$$Z = 11$$

$$x_1 = 4$$

$$x_2 = 5$$

$$x_3 = 0$$

$$\begin{aligned} \text{Minimize} \quad & 5x + 6y \\ \text{subject to} \quad & x + y \geq 2 \\ & 4x + y \geq 4 \\ & x \geq 0,\ y \geq 0 \end{aligned}$$

Canonical Form:

$$\begin{aligned} \text{Minimize} \quad & 5x + 6y \\ \text{subject to} \quad & x + y \geq 2 \\ & 4x + y \geq 4 \\ & x \geq 0,\ y \geq 0 \end{aligned}$$

$$\begin{aligned} \text{Minimize} \quad & 5x + 6y \\ \text{subject to} \quad & x + y \geq 2 \\ & 4x + y \geq 4 \\ & x \geq 0,\ y \geq 0 \end{aligned}$$

Standard Form:

$$\begin{aligned} \text{Maximize} \quad & -5x - 6y \\ \text{subject to} \quad & s_1 - x - y = -2 \\ & s_2 - 4x - y = -4 \\ & x \geq 0,\ y \geq 0 \end{aligned}$$

			0	0	-5	-6	
BV	C	B	s_1	s_2	x	y	MR
s_1	0	-2	1	0	-1	-1	
s_2	0	-4	0	1	-4	-1	
		$Z_j - C_j$	0	0	5	6	

			0	0	−5	−6	
BV	C	B	s_1	s_2	x	y	MR
s_1	0	−1	1	$-\frac{1}{4}$	0	$-\frac{3}{4}$	
x	−5	1	0	$-\frac{1}{4}$	1	$\frac{1}{4}$	
		$Z_j - C_j$	0	$\frac{5}{4}$	0	$\frac{19}{4}$	

			0	0	−5	−6	
BV	C	B	s_1	s_2	x	y	MR
s_2	0	4	−4	1	0	3	
x	−5	2	−1	0	1	1	
		$Z_j - C_j$	5	0	0	1	

$$Z = -10$$

$$x = 2$$

$$y = 0$$

$$\begin{aligned} \text{Maximize} \quad & 3x_1 + 5x_2 \\ \text{subject to} \quad & x_1 + x_2 \leq 1 \\ & 2x_1 + 3x_2 \leq 1 \\ & x \geq 0,\ y \geq 0 \end{aligned}$$

Standard Form:

$$\begin{aligned} \text{Maximize} \quad & 3x_1 + 5x_2 \\ \text{subject to} \quad & s_1 + x_1 + x_2 = 1 \\ & s_2 + 2x_1 + 3x_2 = 1 \\ & x \geq 0,\ y \geq 0 \end{aligned}$$

			0	0	3	5	
BV	C	B	s_1	s_2	x_1	x_2	MR
s_1	0	1	1	0	1	1	1
s_2	0	1	0	1	2	3	$\frac{1}{3}$
		$Z_j - C_j$	0	0	−3	−5	

$$-\infty < C_{x1} < \frac{10}{3}$$

$$\frac{9}{2} < C_{x2} < \infty$$

$$\begin{aligned} \text{Maximize} \quad & x + 4y \\ \text{subject to} \quad & 2x + 4y \leq 7 \\ & 5x + 3y \leq 15 \\ & x \geq 0,\ y \geq 0 \end{aligned}$$

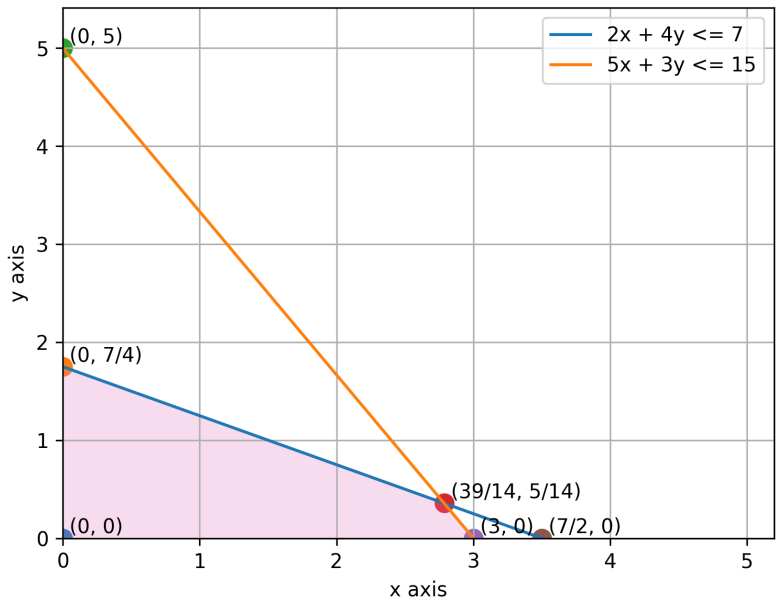
Gauss Elimination:
Echelon Form:

$$\left(\begin{array}{cc|c} 2 & 4 & 7 \\ 0 & -7 & -\frac{5}{2} \end{array}\right)$$

$$\left(\begin{array}{cc|c} 2 & 4 & 7 \\ 0 & -7 & -\frac{5}{2} \end{array}\right)$$

$$x = \frac{39}{14}$$

$$y = \frac{5}{14}$$



$$Z = 7$$

$$x = 0$$

$$y = \frac{7}{4}$$

$$\begin{aligned} \text{Maximize} \quad & x + 4y \\ \text{subject to} \quad & 2x + 4y \leq 7 \\ & 5x + 3y \leq 15 \\ & y \leq 1 \\ & x \geq 0,\ y \geq 0 \end{aligned}$$

Gauss Elimination:
Echelon Form:

$$\left(\begin{array}{cc|c} 2 & 4 & 7 \\ 0 & -7 & -\frac{5}{2} \end{array}\right)$$

$$\left(\begin{array}{cc|c} 2 & 4 & 7 \\ 0 & -7 & -\frac{5}{2} \end{array}\right)$$

$$x = \frac{39}{14}$$

$$y = \frac{5}{14}$$

Gauss Elimination:
Echelon Form:

$$\left(\begin{array}{cc|c} 2 & 4 & 7 \\ 0 & 1 & 1 \end{array}\right)$$

$$\left(\begin{array}{cc|c} 2 & 4 & 7 \\ 0 & 1 & 1 \end{array}\right)$$

$$x = \frac{3}{2}$$

$$y = 1$$

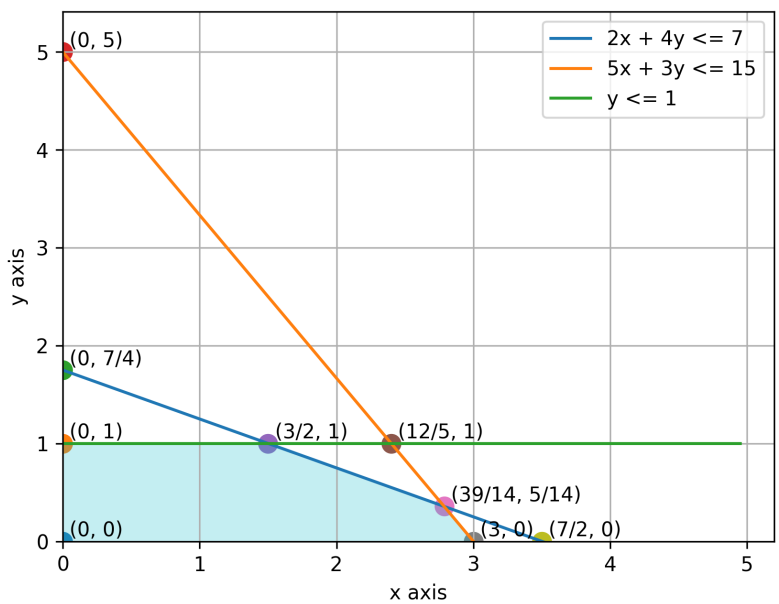
Gauss Elimination:
Echelon Form:

$$\left(\begin{array}{cc|c} 5 & 3 & 15 \\ 0 & 1 & 1 \end{array}\right)$$

$$\left(\begin{array}{cc|c} 5 & 3 & 15 \\ 0 & 1 & 1 \end{array}\right)$$

$$x = \frac{12}{5}$$

$$y = 1$$



$$Z = \frac{11}{2}$$

$$x = \frac{3}{2}$$

$$y = 1$$

$$\begin{aligned} \text{Maximize} \quad & x + 4y \\ \text{subject to} \quad & 2x + 4y \leq 7 \\ & 5x + 3y \leq 15 \\ & y \geq 2 \\ & x \geq 0,\ y \geq 0 \end{aligned}$$

Gauss Elimination:
Echelon Form:

$$\left(\begin{array}{cc|c} 2 & 4 & 7 \\ 0 & -7 & -\frac{5}{2} \end{array}\right)$$

$$\left(\begin{array}{cc|c} 2 & 4 & 7 \\ 0 & -7 & -\frac{5}{2} \end{array}\right)$$

$$x = \frac{39}{14}$$

$$y = \frac{5}{14}$$

Gauss Elimination:
Echelon Form:

$$\left(\begin{array}{cc|c} 2 & 4 & 7 \\ 0 & 1 & 2 \end{array}\right)$$

$$\left(\begin{array}{cc|c} 2 & 4 & 7 \\ 0 & 1 & 2 \end{array}\right)$$

$$x = -\frac{1}{2}$$

$$y = 2$$

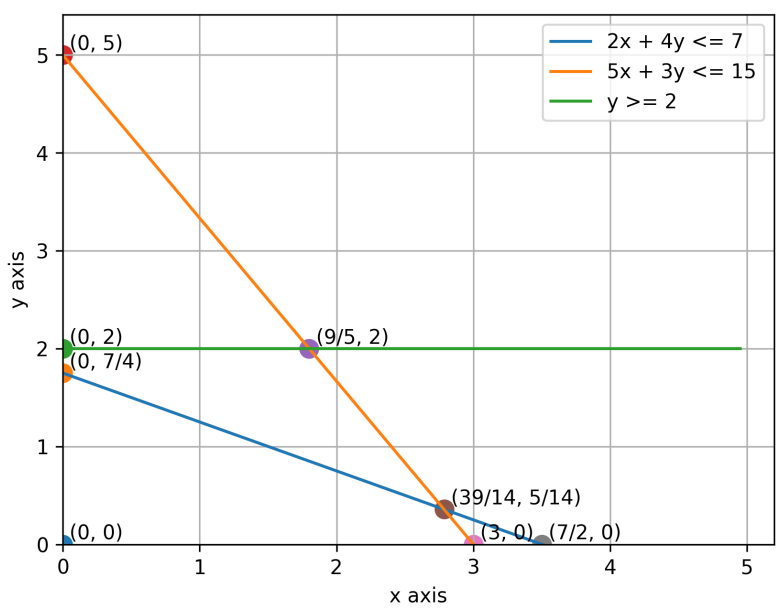
Gauss Elimination:
Echelon Form:

$$\left(\begin{array}{cc|c} 5 & 3 & 15 \\ 0 & 1 & 2 \end{array}\right)$$

$$\left(\begin{array}{cc|c} 5 & 3 & 15 \\ 0 & 1 & 2 \end{array}\right)$$

$$x = \frac{9}{5}$$

$$y = 2$$



Infeasible Solution

$$\begin{aligned} \text{Maximize} \quad & x + 4y \\ \text{subject to} \quad & 2x + 4y \leq 7 \\ & 5x + 3y \leq 15 \\ & y \leq 1 \\ & x \leq 1 \\ & x \geq 0,\ y \geq 0 \end{aligned}$$

Gauss Elimination:
Echelon Form:

$$\left(\begin{array}{cc|c} 2 & 4 & 7 \\ 0 & -7 & -\frac{5}{2} \end{array}\right)$$

$$\left(\begin{array}{cc|c} 2 & 4 & 7 \\ 0 & -7 & -\frac{5}{2} \end{array}\right)$$

$$x = \frac{39}{14}$$

$$y = \frac{5}{14}$$

Gauss Elimination:
Echelon Form:

$$\left(\begin{array}{cc|c} 2 & 4 & 7 \\ 0 & 1 & 1 \end{array}\right)$$

$$\left(\begin{array}{cc|c} 2 & 4 & 7 \\ 0 & 1 & 1 \end{array}\right)$$

$$x = \frac{3}{2}$$

$$y = 1$$

Gauss Elimination:
Echelon Form:

$$\left(\begin{array}{cc|c} 2 & 4 & 7 \\ 0 & -2 & -\frac{5}{2} \end{array}\right)$$

$$\left(\begin{array}{cc|c} 2 & 4 & 7 \\ 0 & -2 & -\frac{5}{2} \end{array}\right)$$

$$x = 1$$

$$y = \frac{5}{4}$$

Gauss Elimination:
Echelon Form:

$$\left(\begin{array}{cc|c} 5 & 3 & 15 \\ 0 & 1 & 1 \end{array}\right)$$

$$\left(\begin{array}{cc|c} 5 & 3 & 15 \\ 0 & 1 & 1 \end{array}\right)$$

$$x = \frac{12}{5}$$

$$y = 1$$

Gauss Elimination:
Echelon Form:

$$\left(\begin{array}{cc|c} 5 & 3 & 15 \\ 0 & -\frac{3}{5} & -2 \end{array}\right)$$

$$\left(\begin{array}{cc|c} 5 & 3 & 15 \\ 0 & -\frac{3}{5} & -2 \end{array}\right)$$

$$x = 1$$

$$y = \frac{10}{3}$$

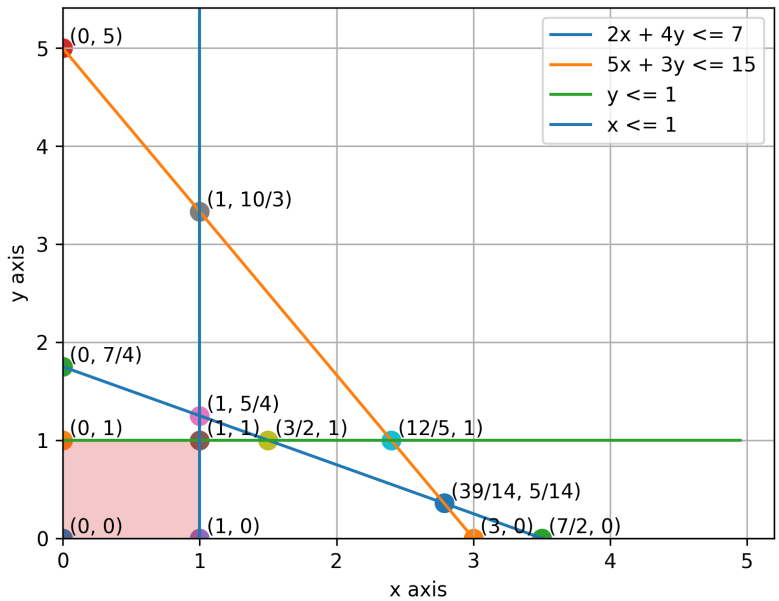
Gauss Elimination:
Echelon Form:

$$\left(\begin{array}{cc|c} 1 & 0 & 1 \\ 0 & 1 & 1 \end{array}\right)$$

$$\left(\begin{array}{cc|c} 1 & 0 & 1 \\ 0 & 1 & 1 \end{array}\right)$$

$$x = 1$$

$$y = 1$$



$$Z = 5$$

$$x = 1$$

$$y = 1$$

$$\begin{aligned} \text{Maximize} \quad & x + 4y \\ \text{subject to} \quad & 2x + 4y \leq 7 \\ & 5x + 3y \leq 15 \\ & y \leq 1 \\ & x \geq 2 \\ & x \geq 0,\ y \geq 0 \end{aligned}$$

Gauss Elimination:
Echelon Form:

$$\left(\begin{array}{cc|c} 2 & 4 & 7 \\ 0 & -7 & -\frac{5}{2} \end{array}\right)$$

$$\left(\begin{array}{cc|c} 2 & 4 & 7 \\ 0 & -7 & -\frac{5}{2} \end{array}\right)$$

$$x = \frac{39}{14}$$

$$y = \frac{5}{14}$$

Gauss Elimination:
Echelon Form:

$$\left(\begin{array}{cc|c} 2 & 4 & 7 \\ 0 & 1 & 1 \end{array}\right)$$

$$\left(\begin{array}{cc|c} 2 & 4 & 7 \\ 0 & 1 & 1 \end{array}\right)$$

$$x = \frac{3}{2}$$

$$y = 1$$

Gauss Elimination:
Echelon Form:

$$\left(\begin{array}{cc|c} 2 & 4 & 7 \\ 0 & -2 & -\frac{3}{2} \end{array}\right)$$

$$\left(\begin{array}{cc|c} 2 & 4 & 7 \\ 0 & -2 & -\frac{3}{2} \end{array}\right)$$

$$x = 2$$

$$y = \frac{3}{4}$$

Gauss Elimination:
Echelon Form:

$$\left(\begin{array}{cc|c} 5 & 3 & 15 \\ 0 & 1 & 1 \end{array}\right)$$

$$\left(\begin{array}{cc|c} 5 & 3 & 15 \\ 0 & 1 & 1 \end{array}\right)$$

$$x = \frac{12}{5}$$

$$y = 1$$

Gauss Elimination:
Echelon Form:

$$\left(\begin{array}{cc|c} 5 & 3 & 15 \\ 0 & -\frac{3}{5} & -1 \end{array}\right)$$

$$\left(\begin{array}{cc|c} 5 & 3 & 15 \\ 0 & -\frac{3}{5} & -1 \end{array}\right)$$

$$x = 2$$

$$y = \frac{5}{3}$$

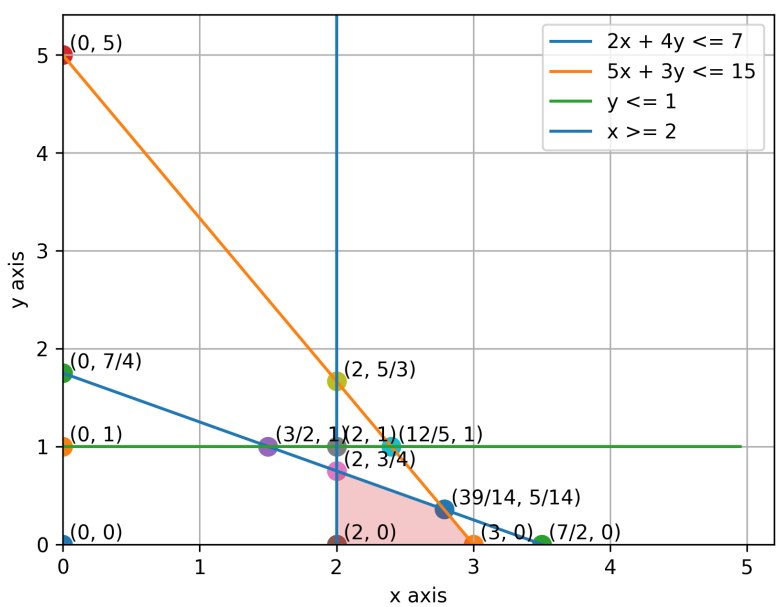
Gauss Elimination:
Echelon Form:

$$\left(\begin{array}{cc|c} 1 & 0 & 2 \\ 0 & 1 & 1 \end{array}\right)$$

$$\left(\begin{array}{cc|c} 1 & 0 & 2 \\ 0 & 1 & 1 \end{array}\right)$$

$$x = 2$$

$$y = 1$$



$$Z = 5$$

$$x = 2$$

$$y = \frac{3}{4}$$

$$\begin{aligned} \text{Maximize} \quad & x + 4y \\ \text{subject to} \quad & 2x + 4y \leq 7 \\ & 5x + 3y \leq 15 \\ & y \leq 1 \\ & x \geq 2 \\ & y \leq 0 \\ & x \geq 0,\ y \geq 0 \end{aligned}$$

Gauss Elimination:
Echelon Form:

$$\left(\begin{array}{cc|c} 2 & 4 & 7 \\ 0 & -7 & -\frac{5}{2} \end{array}\right)$$

$$\left(\begin{array}{cc|c} 2 & 4 & 7 \\ 0 & -7 & -\frac{5}{2} \end{array}\right)$$

$$x = \frac{39}{14}$$

$$y = \frac{5}{14}$$

Gauss Elimination:
Echelon Form:

$$\left(\begin{array}{cc|c} 2 & 4 & 7 \\ 0 & 1 & 1 \end{array}\right)$$

$$\left(\begin{array}{cc|c} 2 & 4 & 7 \\ 0 & 1 & 1 \end{array}\right)$$

$$x = \frac{3}{2}$$

$$y = 1$$

Gauss Elimination:
Echelon Form:

$$\left(\begin{array}{cc|c} 2 & 4 & 7 \\ 0 & -2 & -\frac{3}{2} \end{array}\right)$$

$$\left(\begin{array}{cc|c} 2 & 4 & 7 \\ 0 & -2 & -\frac{3}{2} \end{array}\right)$$

$$x = 2$$

$$y = \frac{3}{4}$$

Gauss Elimination:
Echelon Form:

$$\left(\begin{array}{cc|c} 2 & 4 & 7 \\ 0 & 1 & 0 \end{array}\right)$$

$$\left(\begin{array}{cc|c} 2 & 4 & 7 \\ 0 & 1 & 0 \end{array}\right)$$

$$x = \frac{7}{2}$$

$$y = 0$$

Gauss Elimination:
Echelon Form:

$$\left(\begin{array}{cc|c} 5 & 3 & 15 \\ 0 & 1 & 1 \end{array}\right)$$

$$\left(\begin{array}{cc|c} 5 & 3 & 15 \\ 0 & 1 & 1 \end{array}\right)$$

$$x = \frac{12}{5}$$

$$y = 1$$

Gauss Elimination:
Echelon Form:

$$\left(\begin{array}{cc|c} 5 & 3 & 15 \\ 0 & -\frac{3}{5} & -1 \end{array}\right)$$

$$\left(\begin{array}{cc|c} 5 & 3 & 15 \\ 0 & -\frac{3}{5} & -1 \end{array}\right)$$

$$x = 2$$

$$y = \frac{5}{3}$$

Gauss Elimination:
Echelon Form:

$$\left(\begin{array}{cc|c} 5 & 3 & 15 \\ 0 & 1 & 0 \end{array}\right)$$

$$\left(\begin{array}{cc|c} 5 & 3 & 15 \\ 0 & 1 & 0 \end{array}\right)$$

$$x = 3$$

$$y = 0$$

Gauss Elimination:
Echelon Form:

$$\left(\begin{array}{cc|c} 1 & 0 & 2 \\ 0 & 1 & 1 \end{array}\right)$$

$$\left(\begin{array}{cc|c} 1 & 0 & 2 \\ 0 & 1 & 1 \end{array}\right)$$

$$x = 2$$

$$y = 1$$

Gauss Elimination:

Echelon Form:

$$\left(\begin{array}{c|c} 1 & 1 \\ 0 & -1 \end{array}\right)$$

$$\left(\begin{array}{c|c} 1 & 1 \\ 0 & -1 \end{array}\right)$$

No solution

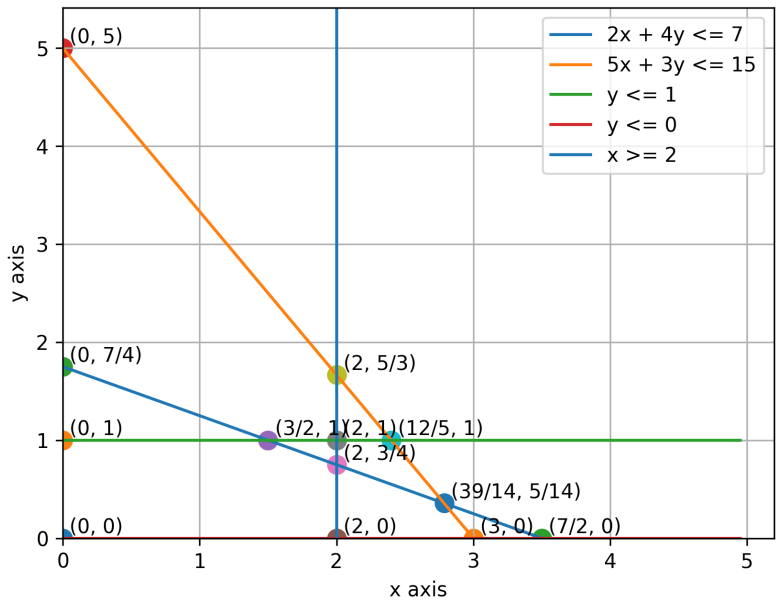
Gauss Elimination:
Echelon Form:

$$\left(\begin{array}{cc|c} 1 & 0 & 2 \\ 0 & 1 & 0 \end{array}\right)$$

$$\left(\begin{array}{cc|c} 1 & 0 & 2 \\ 0 & 1 & 0 \end{array}\right)$$

$$x = 2$$

$$y = 0$$



$$Z = 3$$

$$x = 3$$

$$y = 0$$

$$\begin{aligned} \text{Maximize} \quad & x + 4y \\ \text{subject to} \quad & 2x + 4y \leq 7 \\ & 5x + 3y \leq 15 \\ & y \leq 1 \\ & x \geq 2 \\ & y \geq 1 \\ & x \geq 0,\ y \geq 0 \end{aligned}$$

Gauss Elimination:
Echelon Form:

$$\left(\begin{array}{cc|c} 2 & 4 & 7 \\ 0 & -7 & -\frac{5}{2} \end{array}\right)$$

$$\left(\begin{array}{cc|c} 2 & 4 & 7 \\ 0 & -7 & -\frac{5}{2} \end{array}\right)$$

$$x = \frac{39}{14}$$

$$y = \frac{5}{14}$$

Gauss Elimination:
Echelon Form:

$$\left(\begin{array}{cc|c} 2 & 4 & 7 \\ 0 & 1 & 1 \end{array}\right)$$

$$\left(\begin{array}{cc|c} 2 & 4 & 7 \\ 0 & 1 & 1 \end{array}\right)$$

$$x = \frac{3}{2}$$

$$y = 1$$

Gauss Elimination:
Echelon Form:

$$\left(\begin{array}{cc|c} 2 & 4 & 7 \\ 0 & -2 & -\frac{3}{2} \end{array}\right)$$

$$\left(\begin{array}{cc|c} 2 & 4 & 7 \\ 0 & -2 & -\frac{3}{2} \end{array}\right)$$

$$x = 2$$

$$y = \frac{3}{4}$$

Gauss Elimination:
Echelon Form:

$$\left(\begin{array}{cc|c} 2 & 4 & 7 \\ 0 & 1 & 1 \end{array}\right)$$

$$\left(\begin{array}{cc|c} 2 & 4 & 7 \\ 0 & 1 & 1 \end{array}\right)$$

$$x = \frac{3}{2}$$

$$y = 1$$

Gauss Elimination:
Echelon Form:

$$\left(\begin{array}{cc|c} 5 & 3 & 15 \\ 0 & 1 & 1 \end{array}\right)$$

$$\left(\begin{array}{cc|c} 5 & 3 & 15 \\ 0 & 1 & 1 \end{array}\right)$$

$$x = \frac{12}{5}$$

$$y = 1$$

Gauss Elimination:
Echelon Form:

$$\left(\begin{array}{cc|c} 5 & 3 & 15 \\ 0 & -\frac{3}{5} & -1 \end{array}\right)$$

$$\left(\begin{array}{cc|c} 5 & 3 & 15 \\ 0 & -\frac{3}{5} & -1 \end{array}\right)$$

$$x = 2$$

$$y = \frac{5}{3}$$

Gauss Elimination:
Echelon Form:

$$\left(\begin{array}{cc|c} 5 & 3 & 15 \\ 0 & 1 & 1 \end{array}\right)$$

$$\left(\begin{array}{cc|c} 5 & 3 & 15 \\ 0 & 1 & 1 \end{array}\right)$$

$$x = \frac{12}{5}$$

$$y = 1$$

Gauss Elimination:

Echelon Form:

$$\left(\begin{array}{cc|c} 1 & 0 & 2 \\ 0 & 1 & 1 \end{array}\right)$$

$$\left(\begin{array}{cc|c} 1 & 0 & 2 \\ 0 & 1 & 1 \end{array}\right)$$

$$x = 2$$

$$y = 1$$

Gauss Elimination:
Echelon Form:

$$\left(\begin{array}{c|c} 1 & 1 \\ 0 & 0 \end{array}\right)$$

Infinitely many solutions

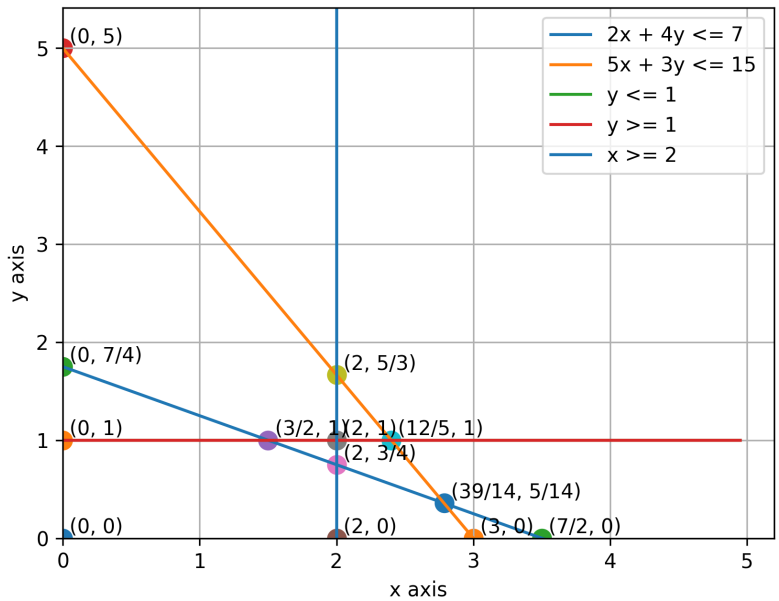
Gauss Elimination:
Echelon Form:

$$\left(\begin{array}{cc|c} 1 & 0 & 2 \\ 0 & 1 & 1 \end{array}\right)$$

$$\left(\begin{array}{cc|c} 1 & 0 & 2 \\ 0 & 1 & 1 \end{array}\right)$$

$$x = 2$$

$$y = 1$$



Infeasible Solution

$$Z = 5$$

$$x = 1$$

$$y = 1$$